

SAPIENZA
UNIVERSITÀ DI ROMA

Linear Programming

Catalogue of LP problems

The linear programming problems used in the examples and exercises of the whole course, cited by their code P0–P86: problems with two and three variables, in four types (A: non-negative right-hand sides; B: dual simplex; C: Phase 0 needed; D: unbounded, infeasible or with multiple optima). For each, the problem and the figure of its feasible region.

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Online version and downloadable material:
fabiofurini.github.io/linear-programming/
github.com/fabiofurini/linear-programming

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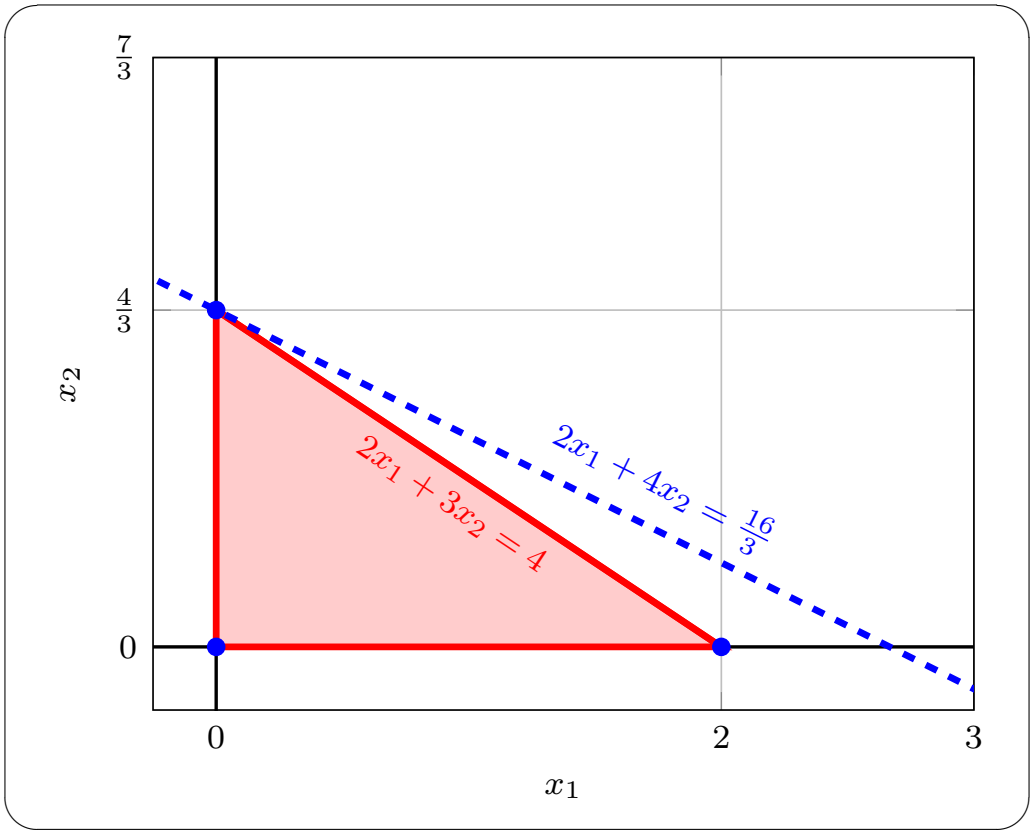
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Chapter 1

LP problems with two variables – Type A

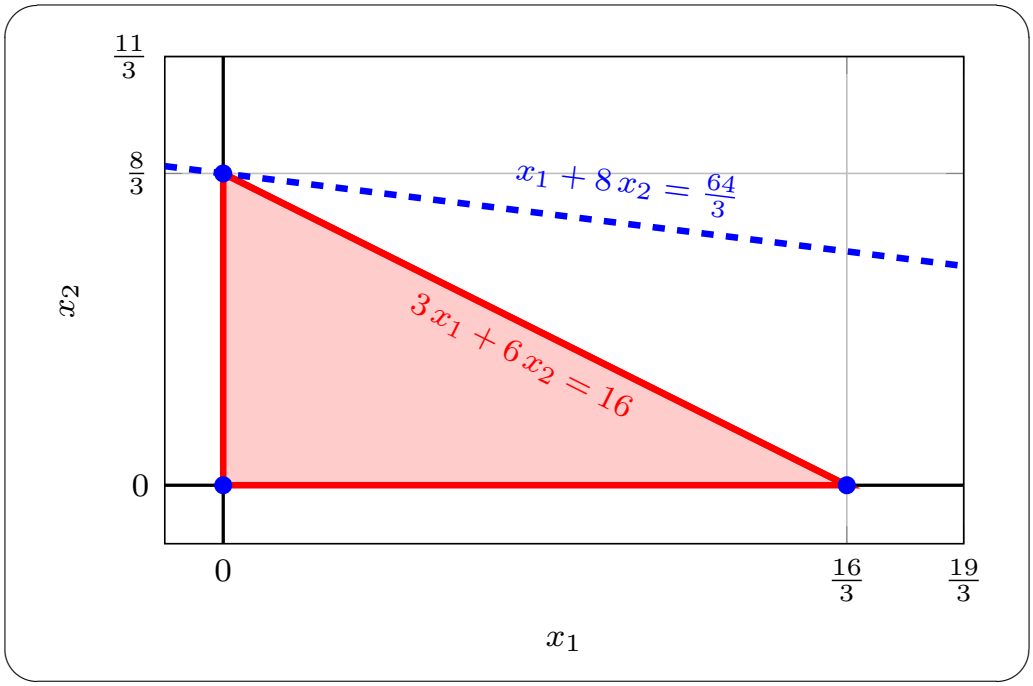
1.1 LP Problem $P0$

$$
\begin{array}{rrcrcrl}
\max & 2 & x_1 & + & 4 & x_2 & \\
\text{subject to} & 2 & x_1 & + & 3 & x_2 & \leq 4 \\
 & & x_1, & & & x_2 & \geq 0
\end{array}
$$



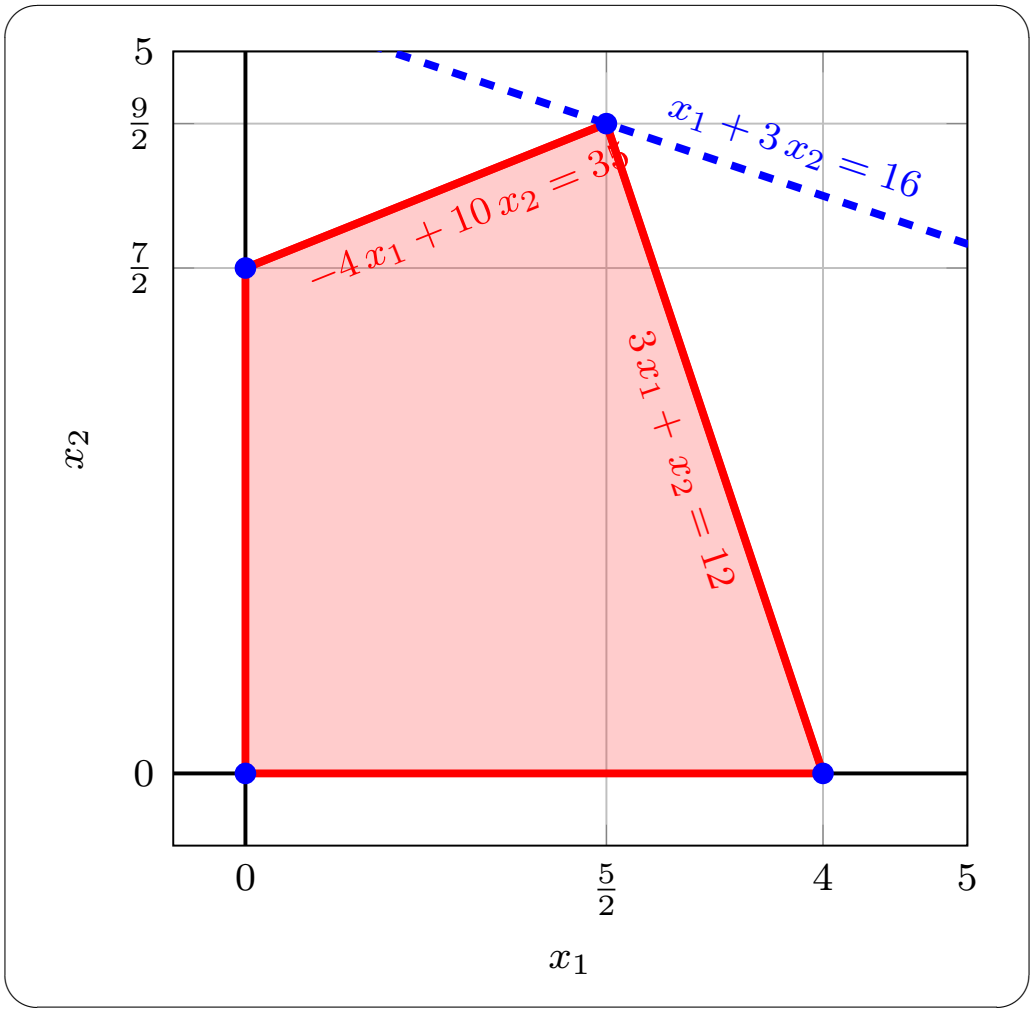
1.2 LP Problem $P1$

$$
\begin{array}{lrcrcl}
\max & x_1 & + & 8\ x_2 & & \\
\text{subject to} & 3\ x_1 & + & 6\ x_2 & \leq & 16 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



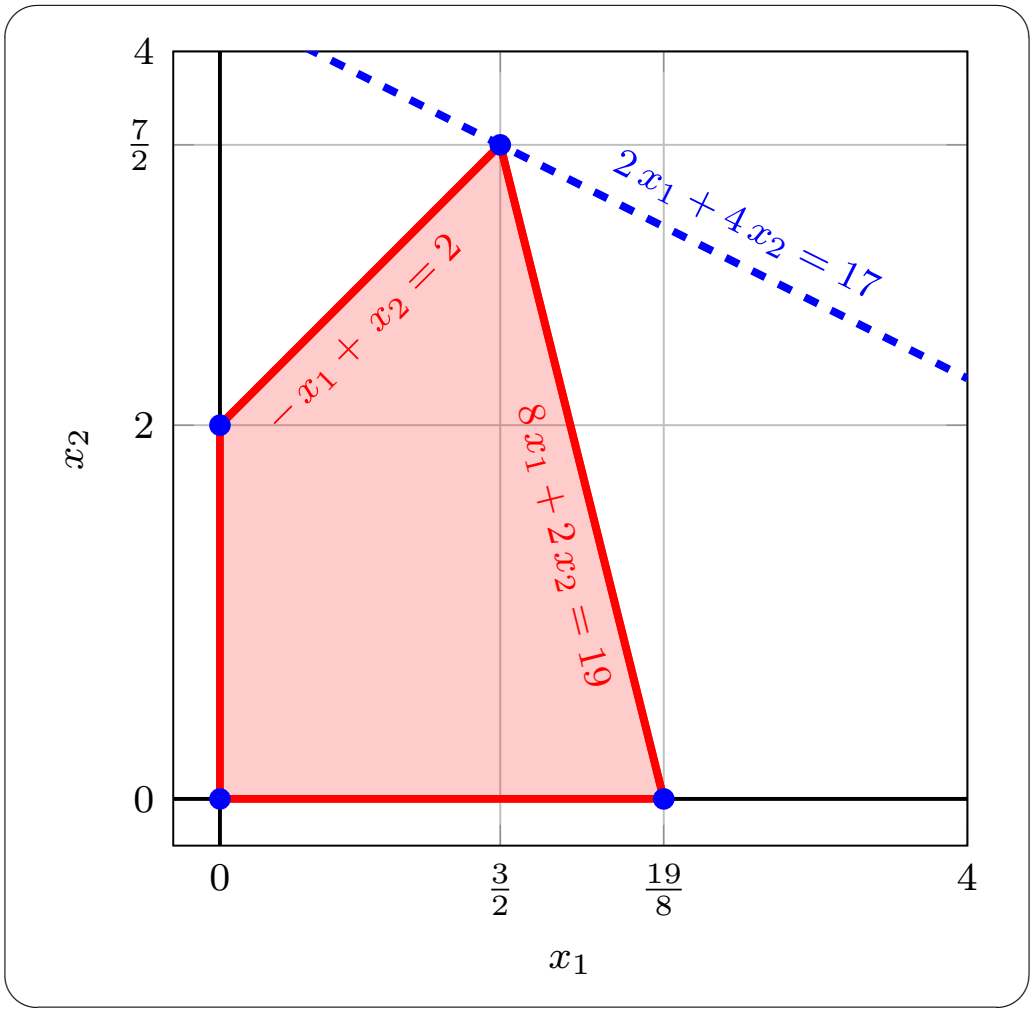
1.3 LP Problem *P2*

$$
\begin{array}{rrcrcrl}
\max & x_1 & + & 3 & x_2 & & \\
\text{subject to} & -4\,x_1 & + & 10 & x_2 & \leq & 35 \\
& 3\,x_1 & + & & x_2 & \leq & 12 \\
& x_1, & & & x_2 & \geq & 0
\end{array}
$$



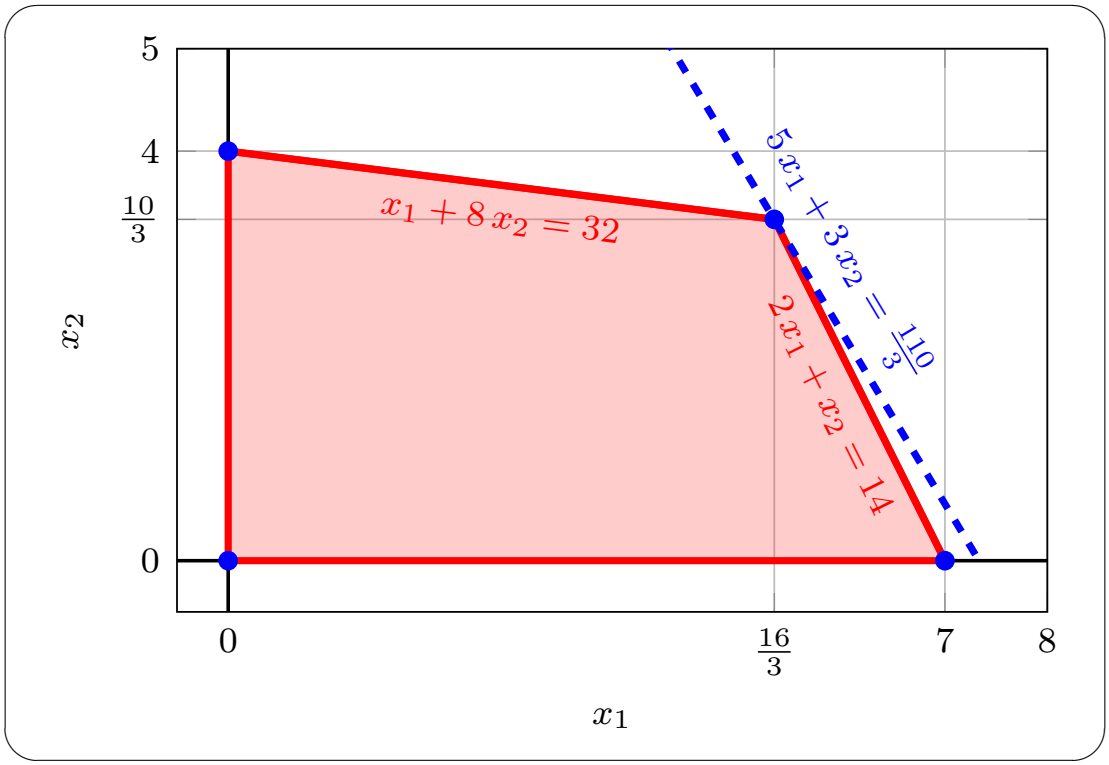
1.4 LP Problem *P3*

$$
\begin{array}{rrcrcrcl}
\max & 2 & x_1 & + & 4 & x_2 & & \\
\text{subject to} & - & x_1 & + & & x_2 & \leq & 2 \\
& 8 & x_1 & + & 2 & x_2 & \leq & 19 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$



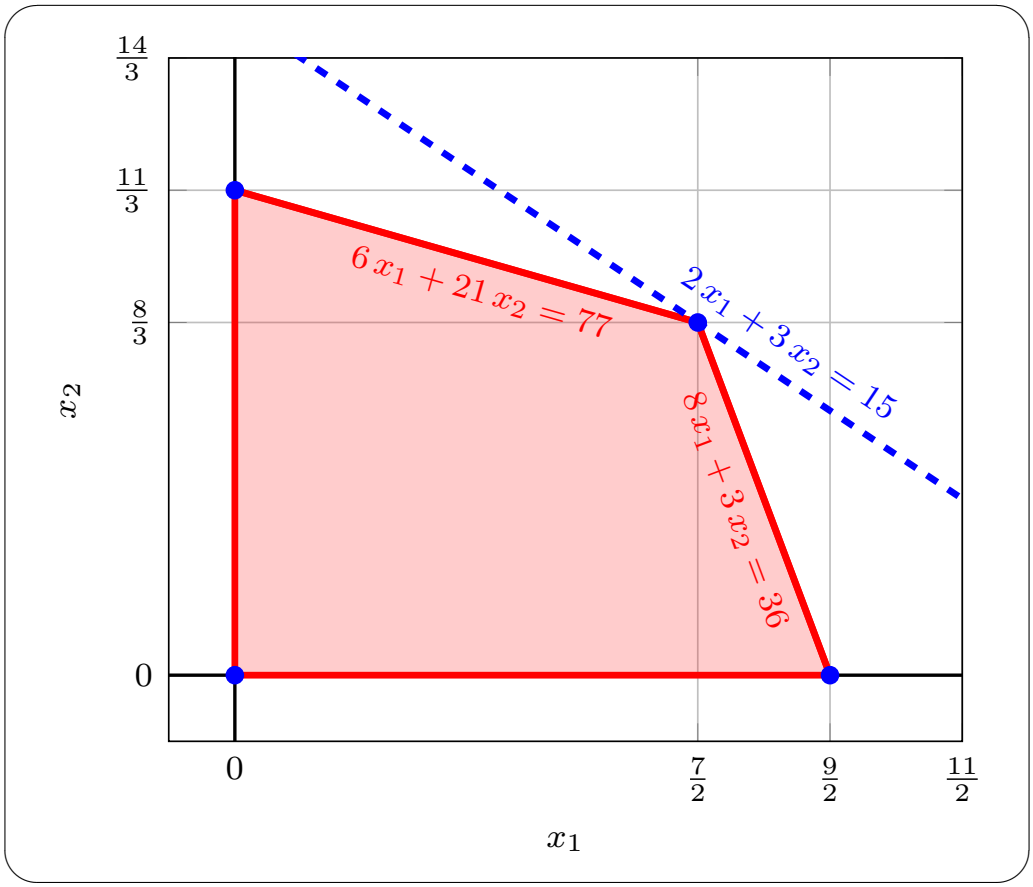
1.5 LP Problem *P4*

$$
\begin{array}{rrcrcrl}
\max & 5\,x_1 & + & 3 & x_2 & & \\
\text{subject to} & 2\,x_1 & + & & x_2 & \leq & 14 \\
& x_1 & + & 8 & x_2 & \leq & 32 \\
& x_1, & & & x_2 & \geq & 0
\end{array}
$$



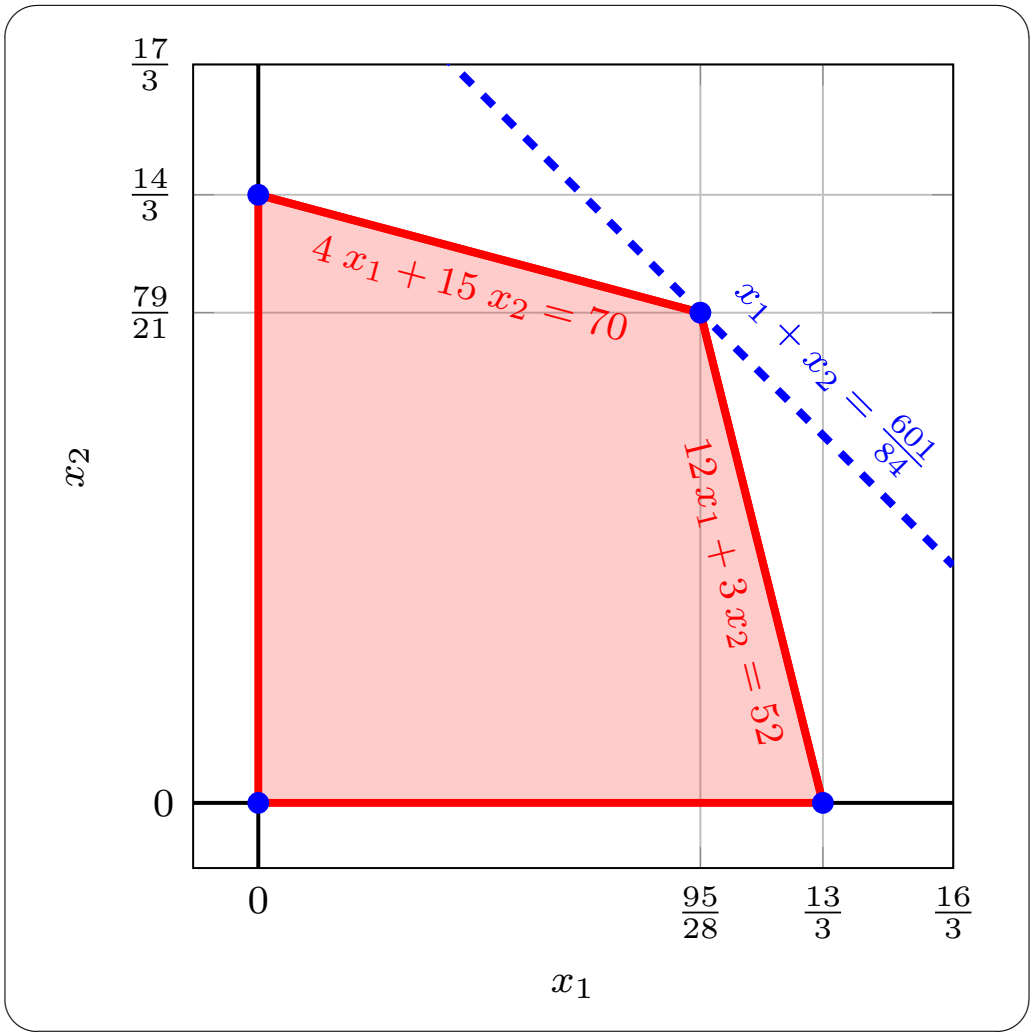
1.6 LP Problem $P5$

$$
\begin{array}{rrcrcl}
\max & 2\,x_1 & + & 3\,x_2 & & \\
\text{subject to} & 6\,x_1 & + & 21\,x_2 & \leq & 77 \\
& 8\,x_1 & + & 3\,x_2 & \leq & 36 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



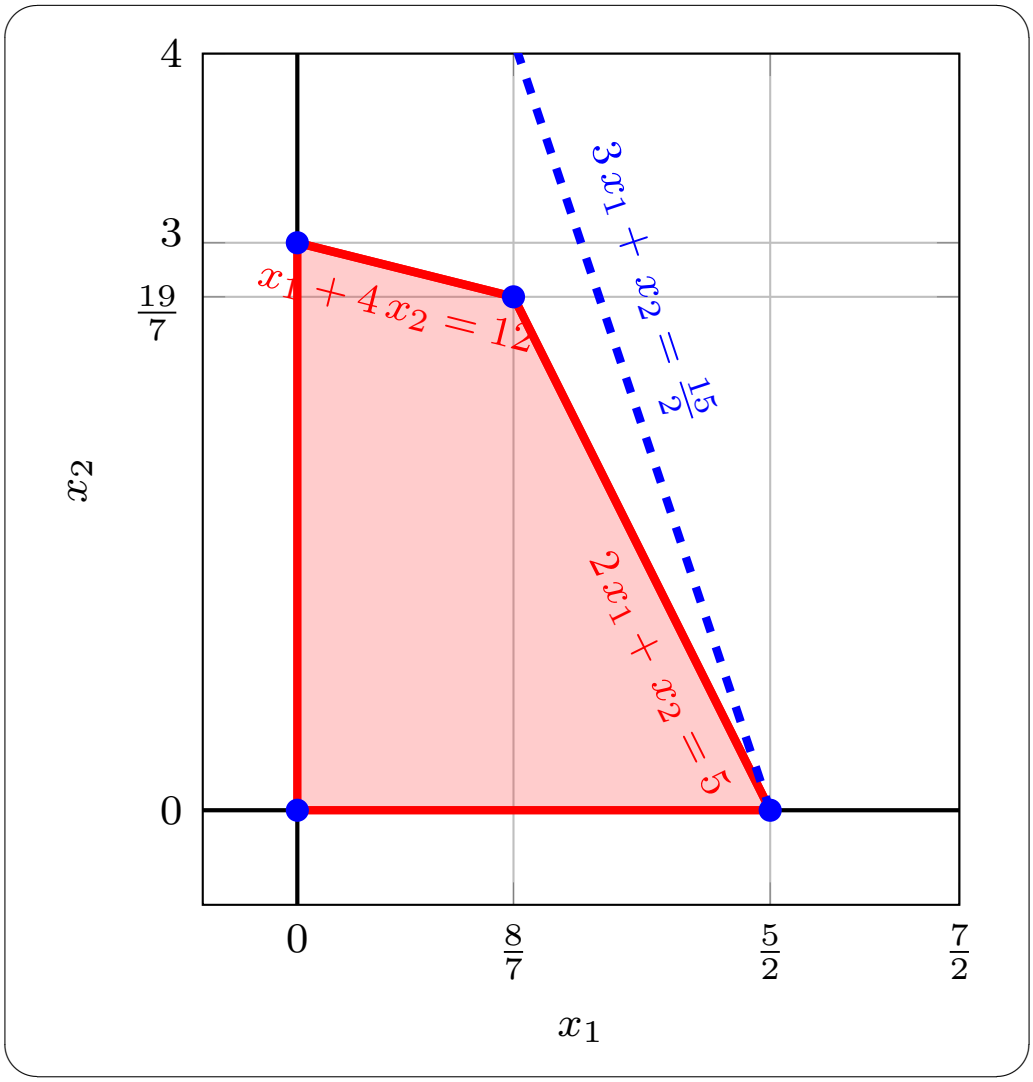
1.7 ILP problem *P6*

$$
\begin{array}{rrcrcrl}
\max & x_1 & + & & x_2 & & \\
\text{subject to} & 4\,x_1 & + & 15\,x_2 & \leq & 70 \\
& 12\,x_1 & + & 3\,x_2 & \leq & 52 \\
& x_1, & & & x_2 & \in & \mathbb{Z}_{\geq 0}
\end{array}
$$



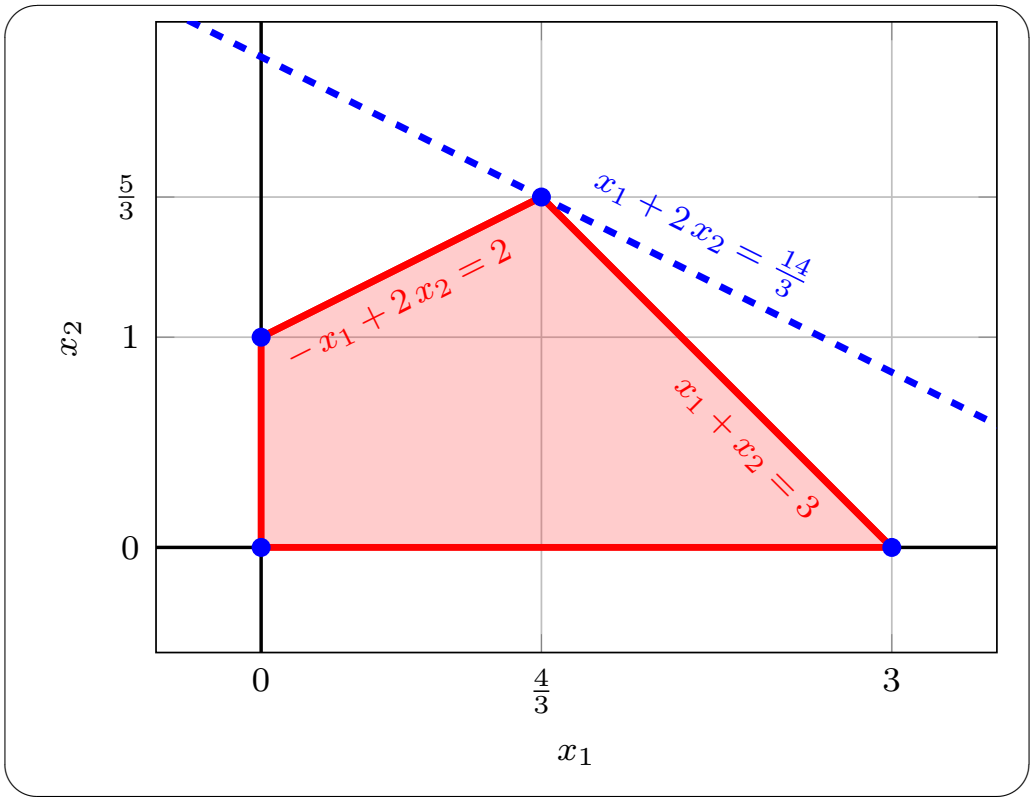
1.8 LP Problem *P7*

$$
\begin{array}{rrcrcrcl}
\max & 3 & x_1 & + & & x_2 & & \\
\text{subject to} & & x_1 & + & 4 & x_2 & \leq & 12 \\
& 2 & x_1 & + & & x_2 & \leq & 5 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$



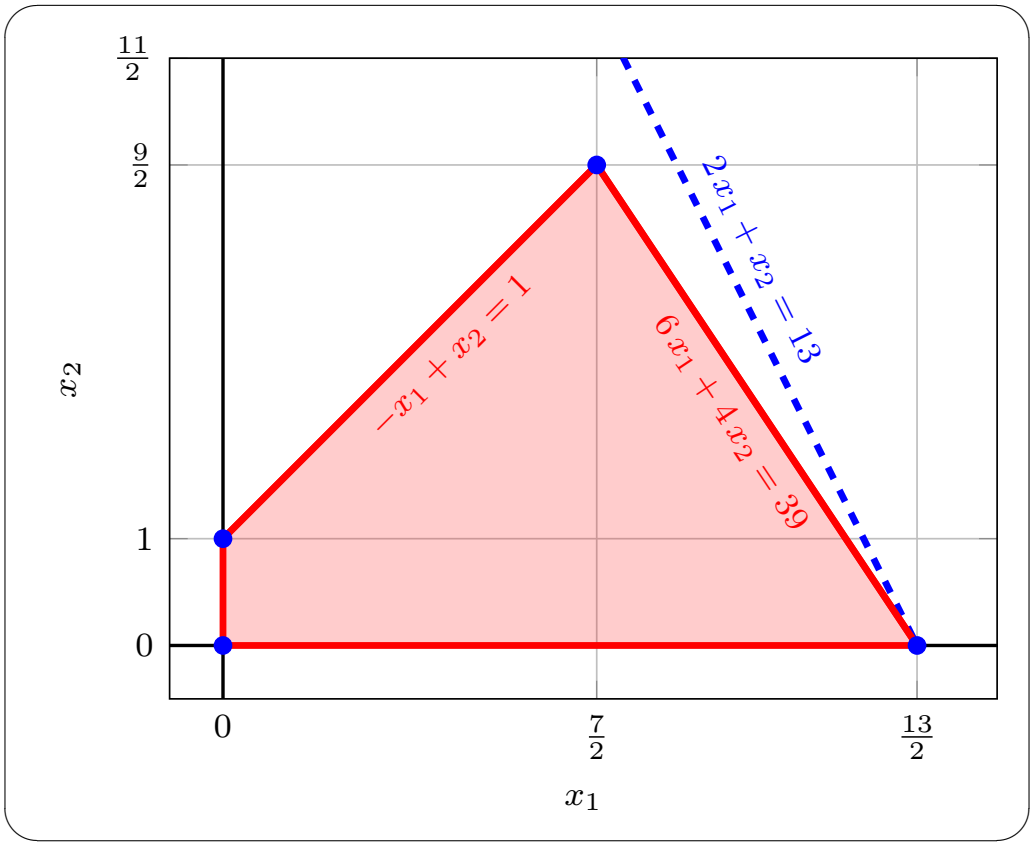
1.9 LP Problem *P8*

$$
\begin{array}{rrcrcl}
\max & x_1 & + & 2\ x_2 & & \\
\text{subject to} & -\ x_1 & + & 2\ x_2 & \leq & 2 \\
& x_1 & + & x_2 & \leq & 3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



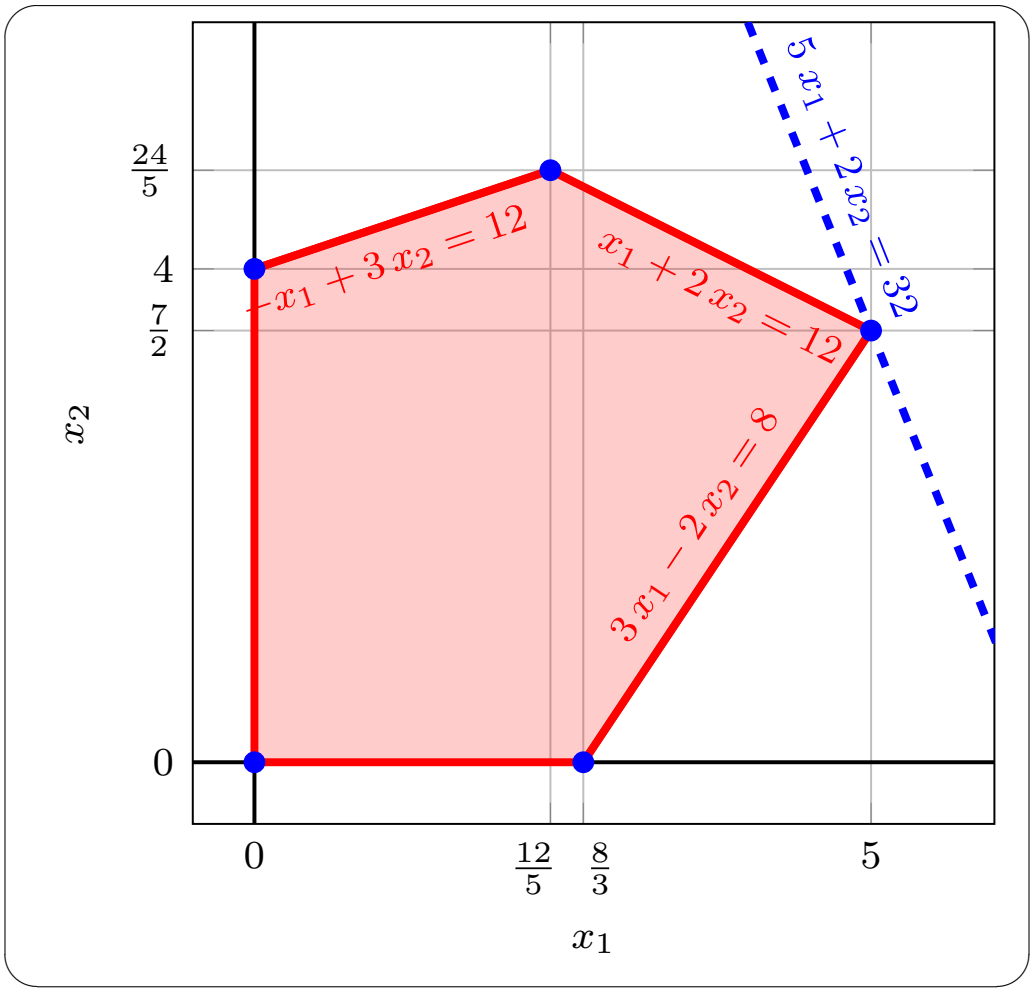
1.10 LP Problem *P9*

$$
\begin{array}{rrrrrrrr}
\max & 2 & x_1 & + & & x_2 & & \\
\text{subject to} & - & x_1 & + & & x_2 & \leq & 1 \\
& 6 & x_1 & + & 4 & x_2 & \leq & 39 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$



1.11 LP Problem *P*10

$$
\begin{array}{rrcrcrcl}
\max & 5 & x_1 & + & 2 & x_2 & & \\
\text{subject to} & - & x_1 & + & 3 & x_2 & \leq & 12 \\
 & & x_1 & + & 2 & x_2 & \leq & 12 \\
 & 3 & x_1 & - & 2 & x_2 & \leq & 8 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$

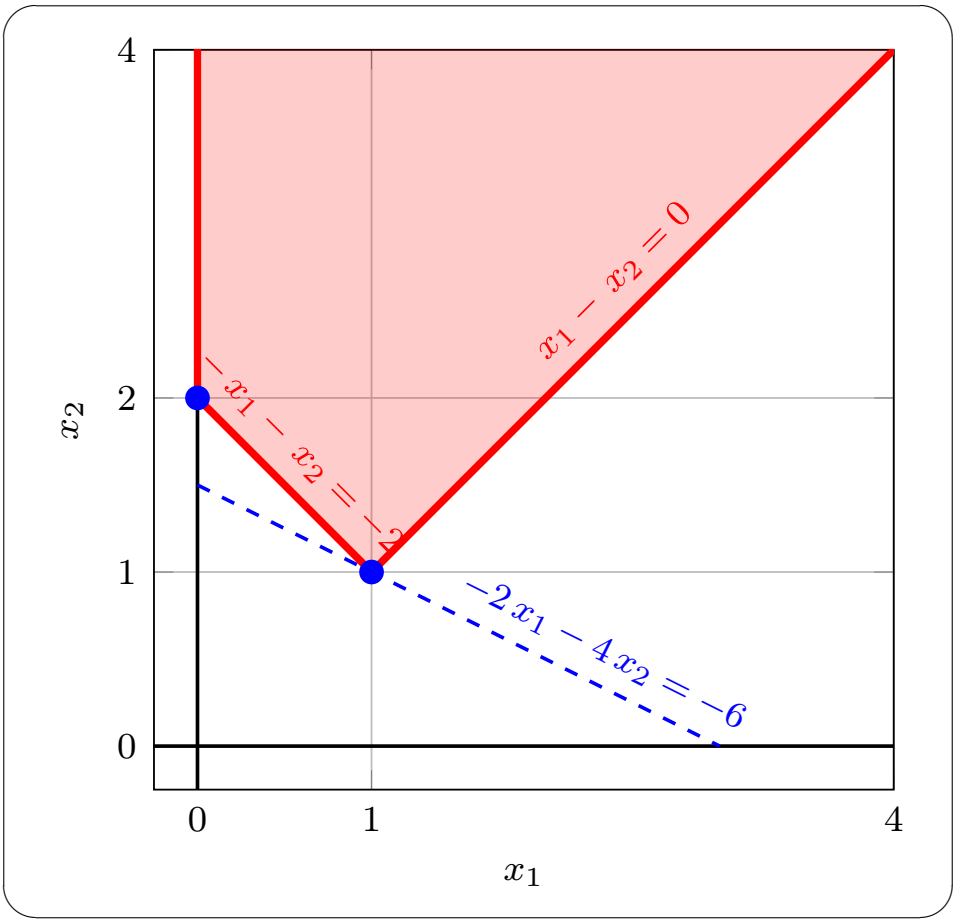


Chapter 2

LP problems with two variables – Type B

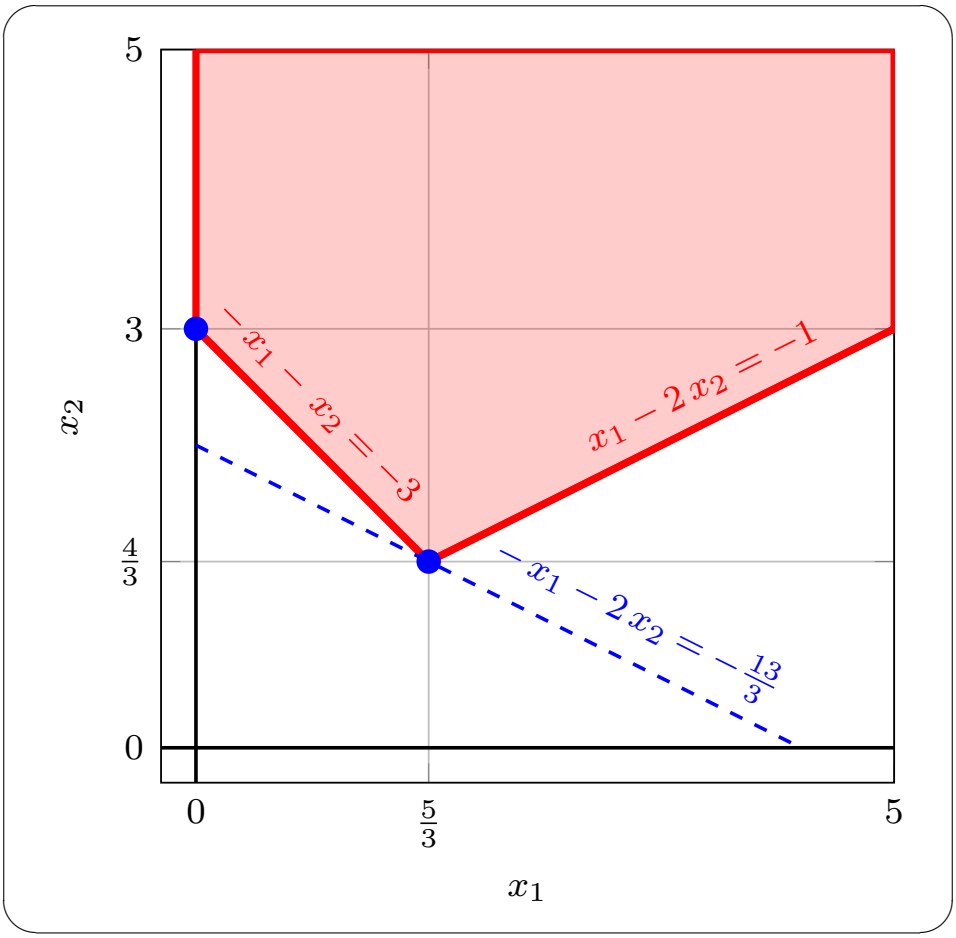
2.1 LP Problem *P11*

$$
\begin{array}{rrcrcr}
\max & -2\,x_1 & - & 4\,x_2 & & \\
\text{subject to} & -\,x_1 & - & x_2 & \leq & -2 \\
& x_1 & - & x_2 & \leq & 0 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



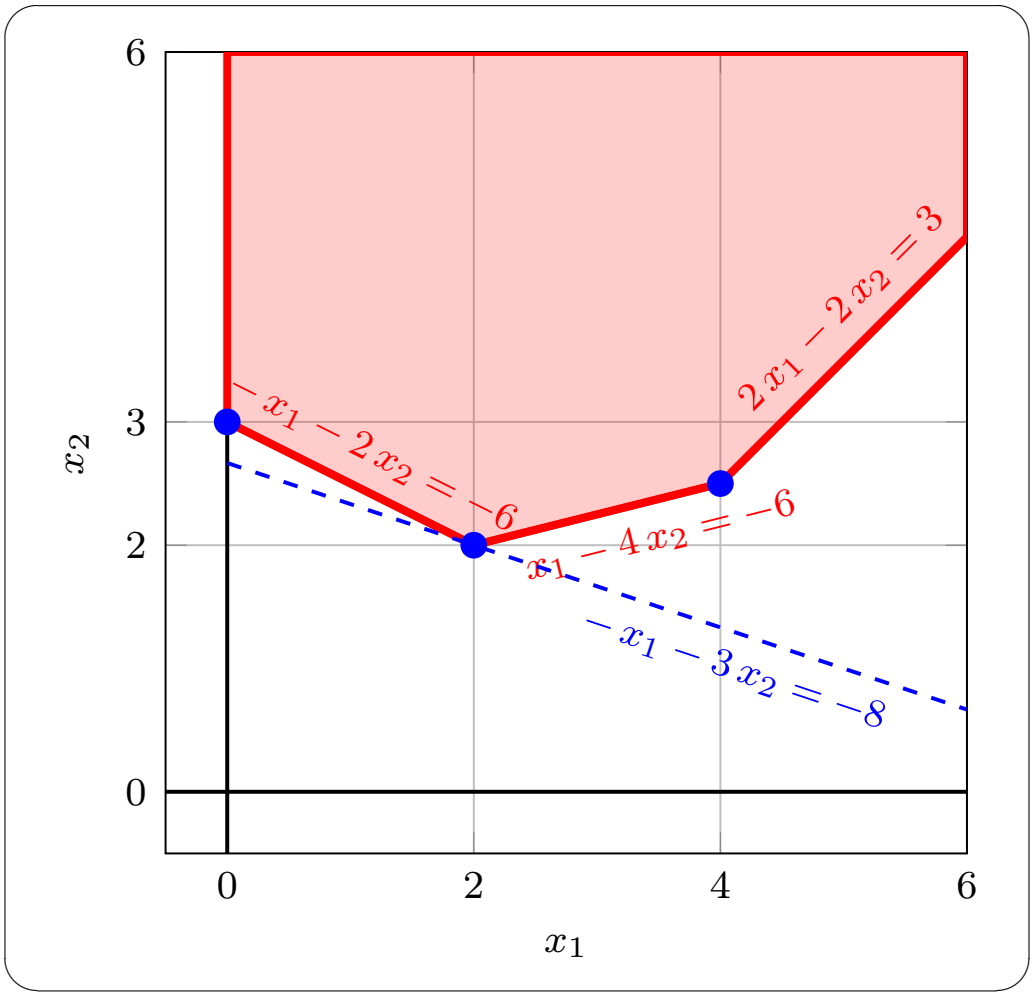
2.2 LP Problem *P12*

$$
\begin{array}{rrrrrr}
\max & -x_1 & - & 2\,x_2 & & \\
\text{subject to} & -x_1 & - & x_2 & \leq & -3 \\
& x_1 & - & 2\,x_2 & \leq & -1 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



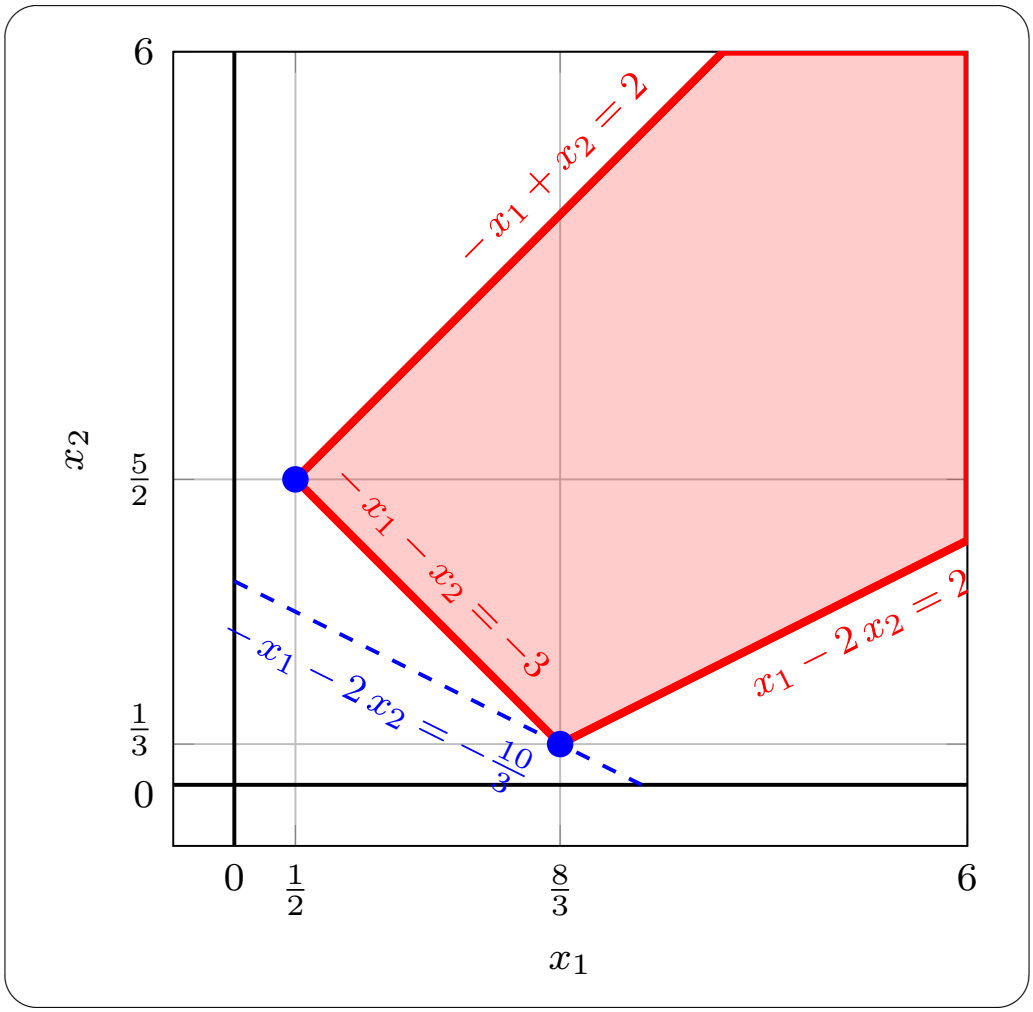
2.3 LP Problem *P13*

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 3 & x_2 & & \\
\text{subject to} & - & x_1 & - & 2 & x_2 & \leq & -6 \\
 & & x_1 & - & 4 & x_2 & \leq & -6 \\
 & 2 & x_1 & - & 2 & x_2 & \leq & 3 \\
 & & x_1, & & & x_2 & \geq & 0
\end{array}
$$



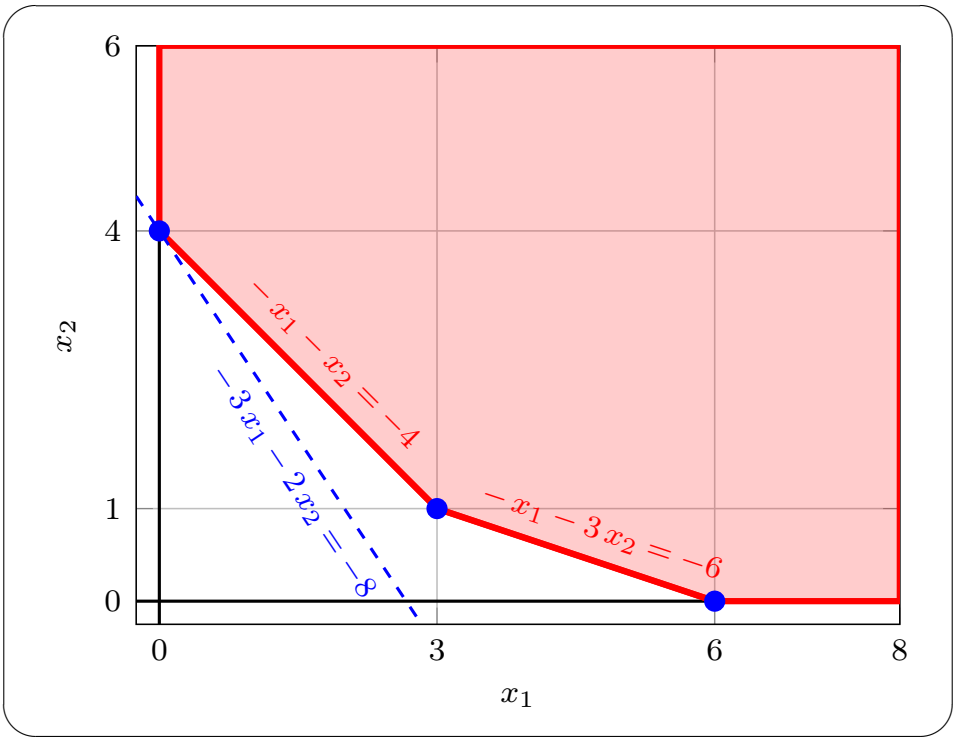
2.4 LP Problem *P14*

$$
\begin{array}{rrrrrrrr}
\max & - & x_1 & - & 2 & x_2 & & \\
\text{subject to} & - & x_1 & + & & x_2 & \leq & 2 \\
& & x_1 & - & 2 & x_2 & \leq & 2 \\
& - & x_1 & - & & x_2 & \leq & -3 \\
& & x_1, & & & x_2 & \geq & 0
\end{array}
$$



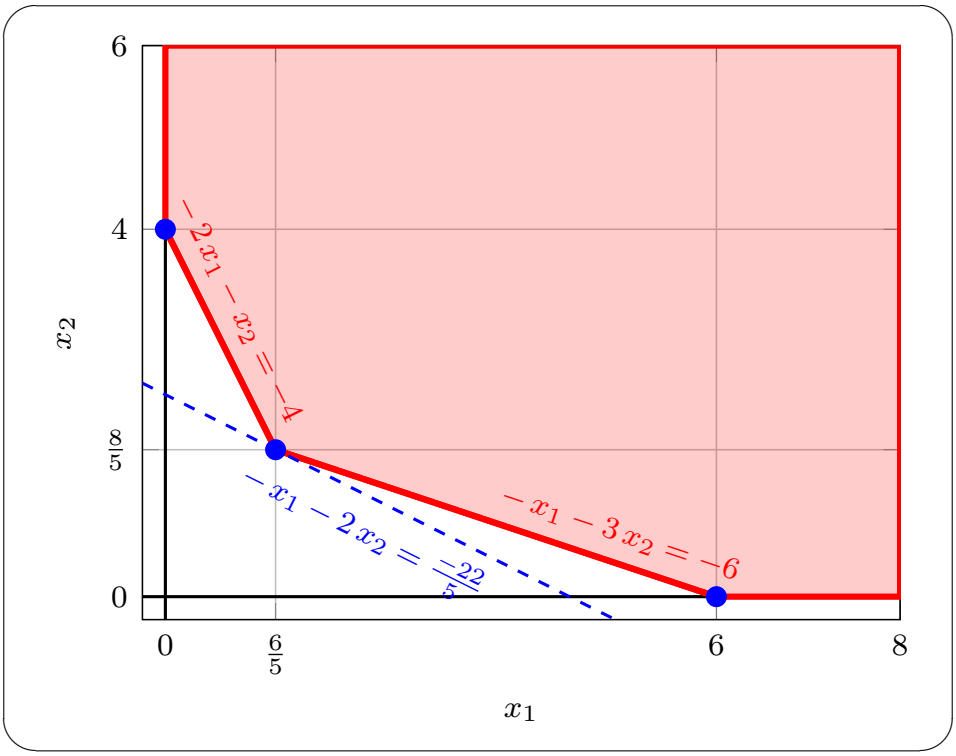
2.5 LP Problem *P15*

$$
\begin{array}{rrcrcr}
\max & -3\,x_1 & - & 2\,x_2 & & \\
\text{subject to} & -\,x_1 & - & x_2 & \leq & -4 \\
& -\,x_1 & - & 3\,x_2 & \leq & -6 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



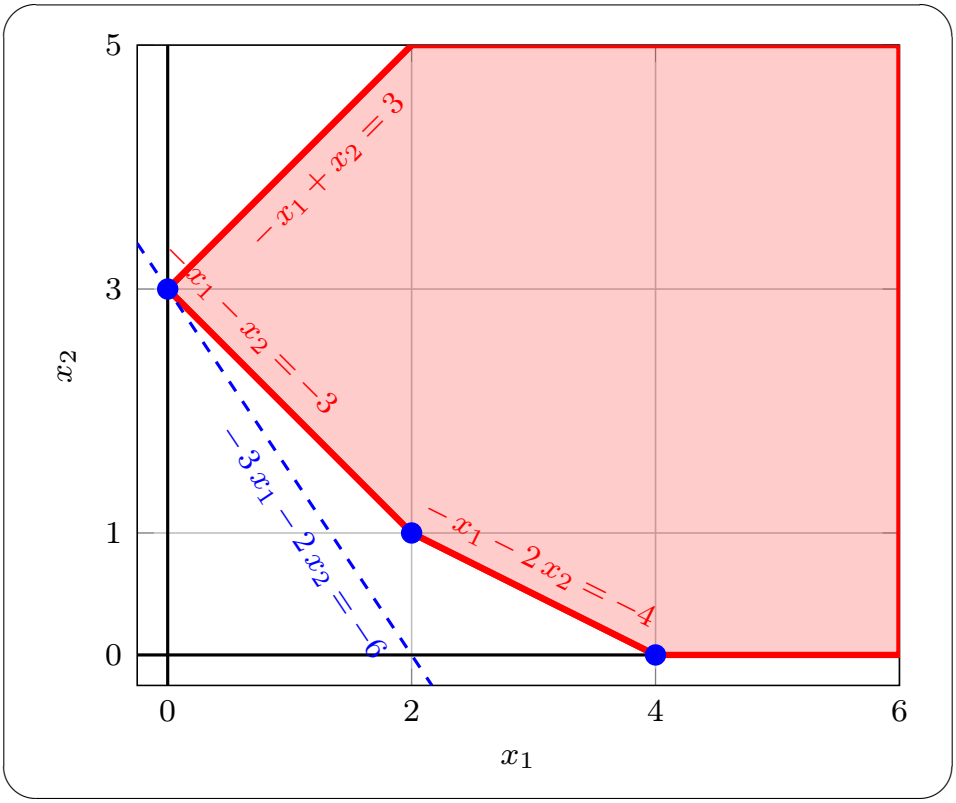
2.6 LP Problem *P16*

$$
\begin{array}{rrcrcr}
\max & -x_1 & - & 2\,x_2 & & \\
\text{subject to} & -2\,x_1 & - & x_2 & \leq & -4 \\
& -x_1 & - & 3\,x_2 & \leq & -6 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



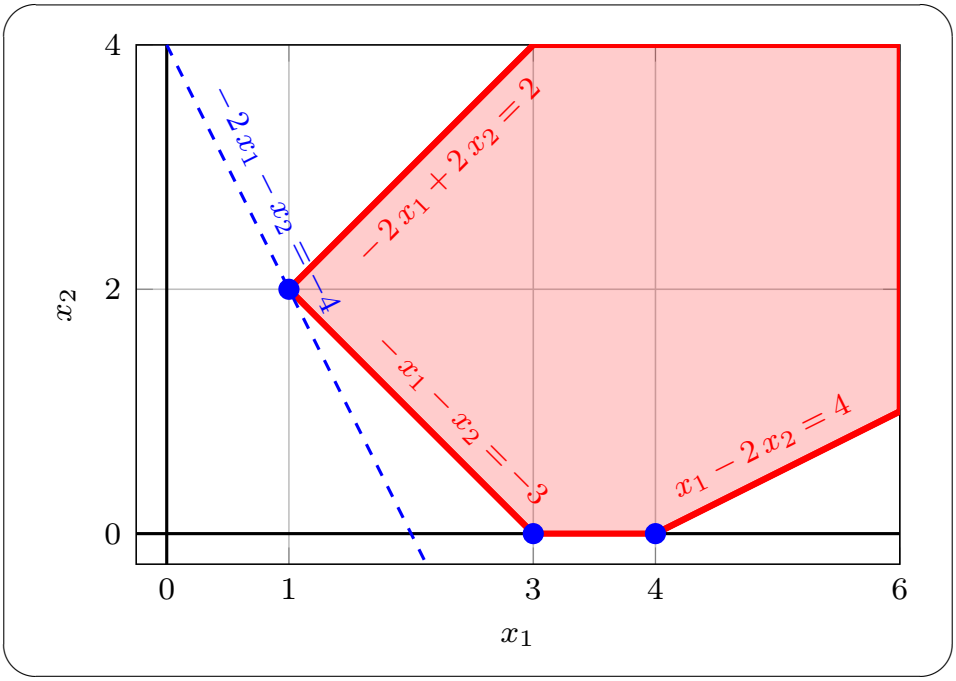
2.7 LP Problem *P17*

$$
\begin{array}{rrrrrr}
\max & -3\,x_1 & - & 2\,x_2 & & \\
\text{subject to} & -\,x_1 & - & x_2 & \leq & -3 \\
& -\,x_1 & + & x_2 & \leq & 3 \\
& -\,x_1 & - & 2\,x_2 & \leq & -4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



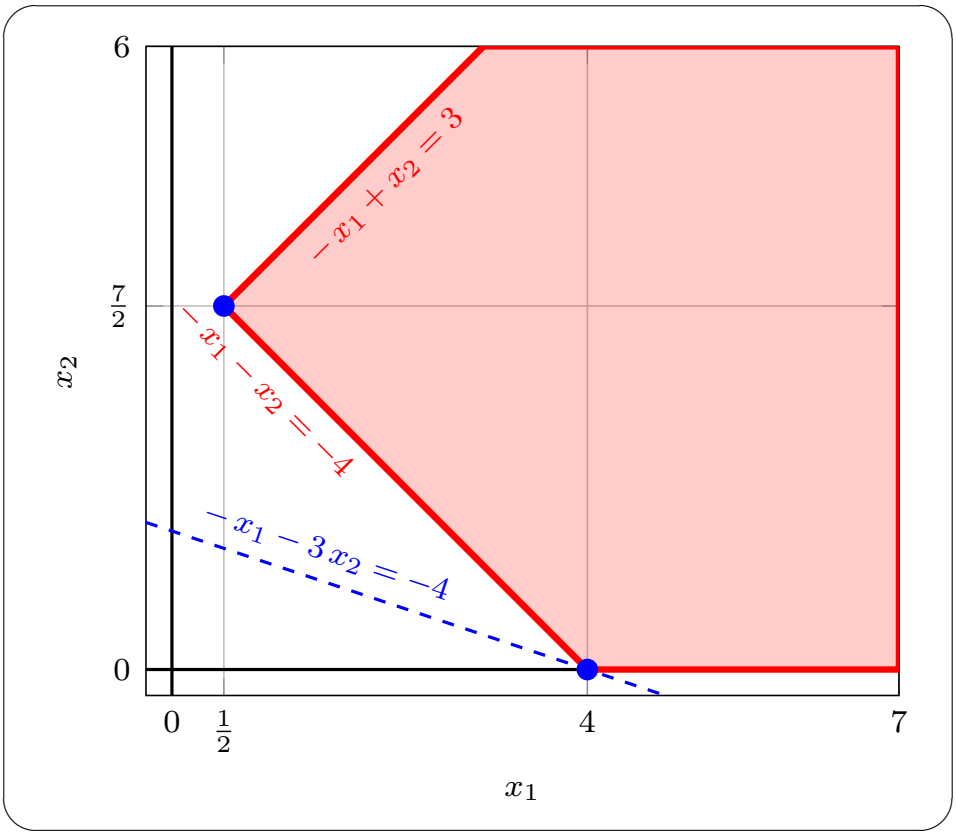
2.8 LP Problem *P18*

$$
\begin{array}{rrcrcr}
\max & -2\,x_1 & - & x_2 & & \\
\text{subject to} & -\,x_1 & - & x_2 & \leq & -3 \\
& x_1 & - & 2\,x_2 & \leq & 4 \\
& -2\,x_1 & + & 2\,x_2 & \leq & 2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



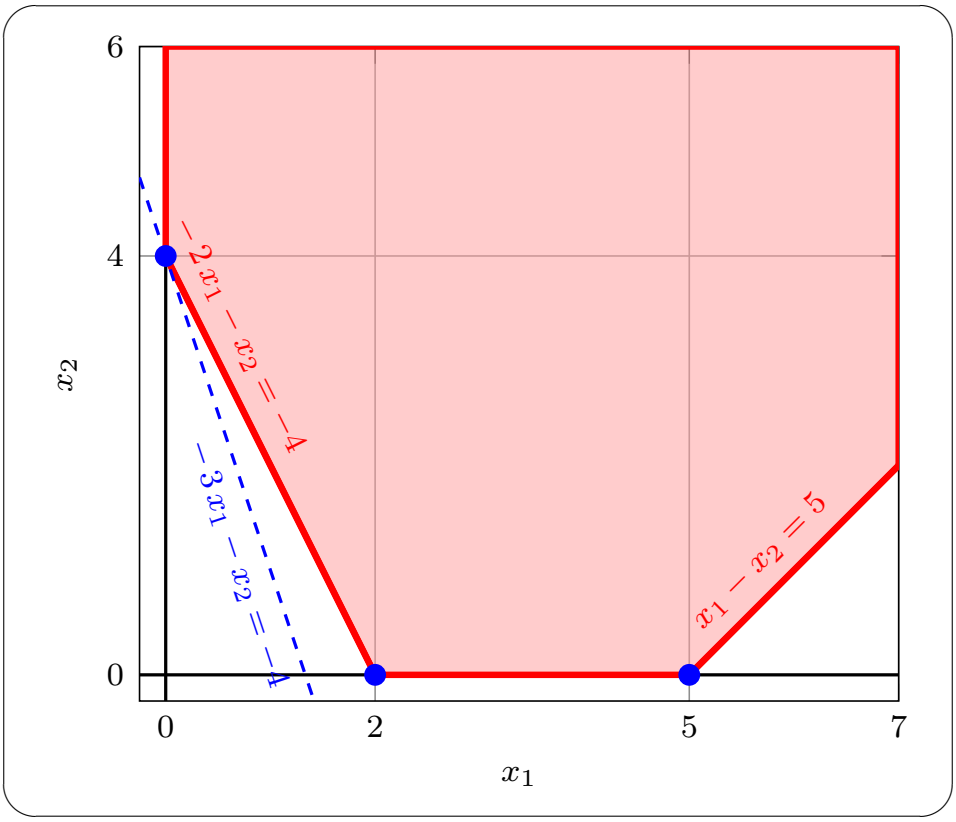
2.9 LP Problem *P19*

$$
\begin{array}{rrrrrr}
\max & -x_1 & - & 3\,x_2 & & \\
\text{subject to} & -x_1 & - & x_2 & \leq & -4 \\
& -x_1 & + & x_2 & \leq & 3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



2.10 LP Problem *P*20

$$
\begin{array}{rrcrcr}
\max & -3\,x_1 & - & x_2 & & \\
\text{subject to} & x_1 & - & x_2 & \leq & 5 \\
& -2\,x_1 & - & x_2 & \leq & -4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$

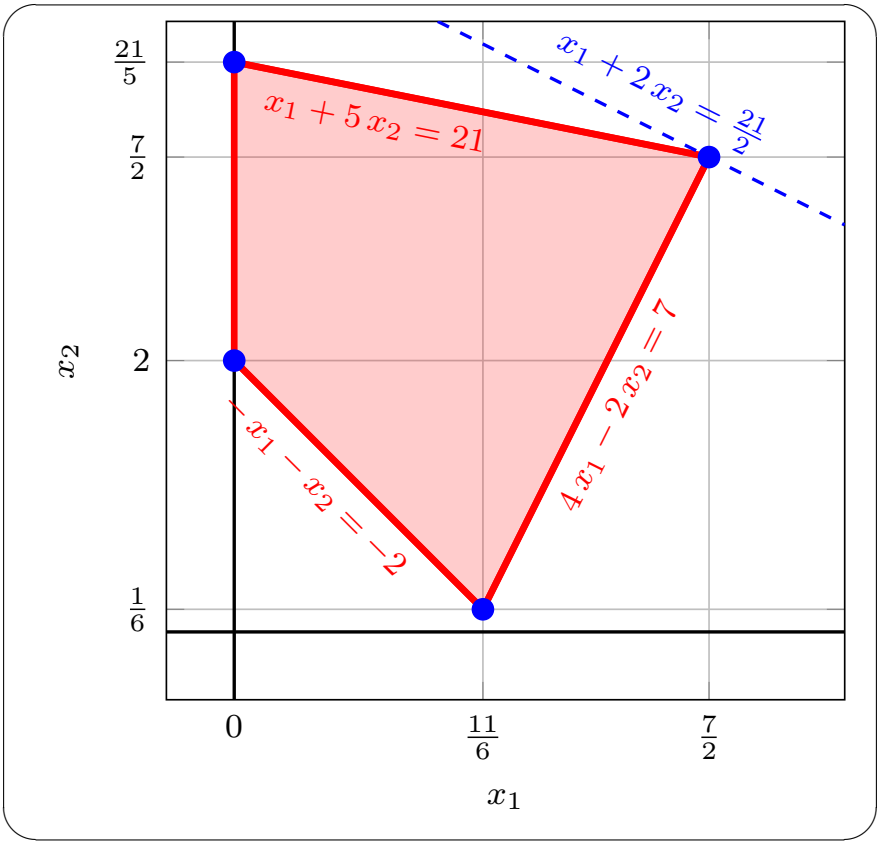


Chapter 3

LP problems with two variables – Type C

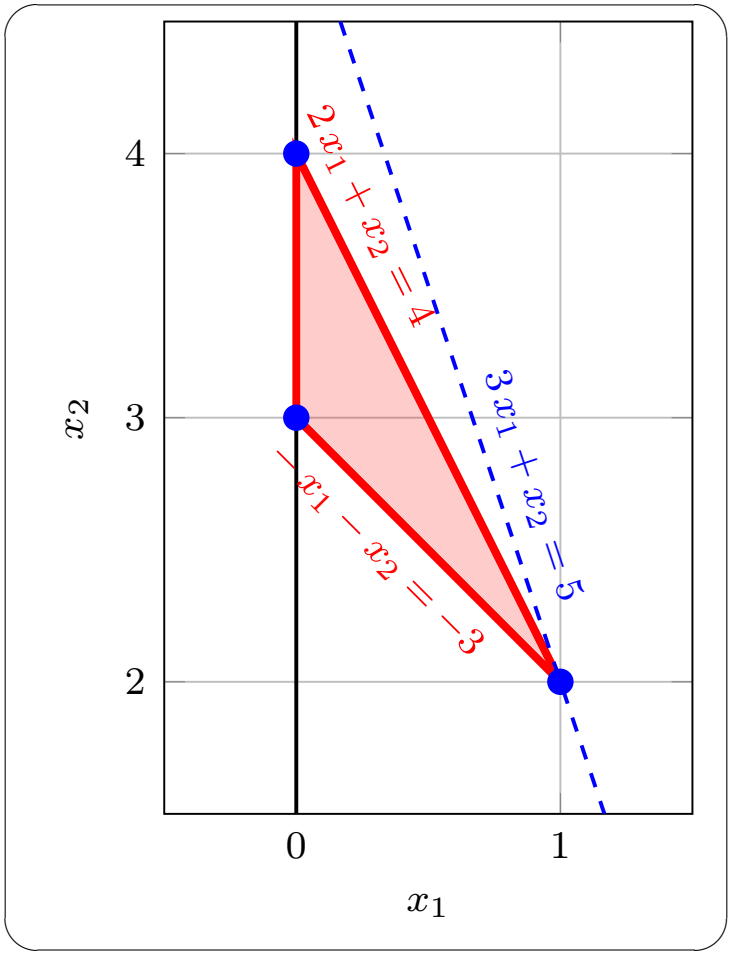
3.1 LP Problem *P21*

$$
\begin{array}{rrrrrr}
\max & x_1 & + & 2\,x_2 & & \\
\text{subject to} & -\,x_1 & - & x_2 & \leq & -2 \\
& 4\,x_1 & - & 2\,x_2 & \leq & 7 \\
& x_1 & + & 5\,x_2 & \leq & 21 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



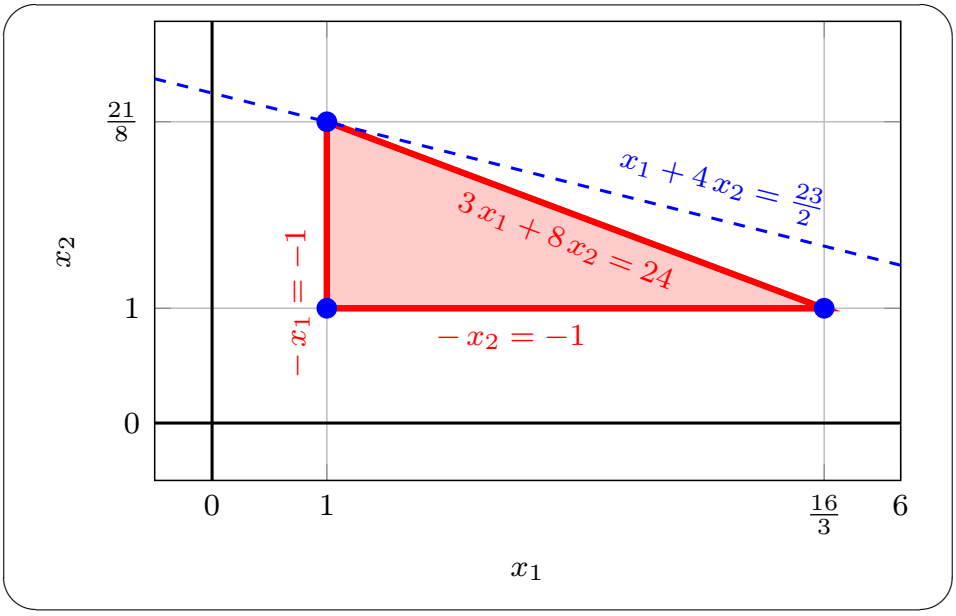
3.2 LP Problem *P22*

$$
\begin{array}{rrcrcr}
\max & 3\,x_1 & + & x_2 & & \\
\text{subject to} & x_1 & - & x_2 & \leq & -1 \\
& -\,x_1 & - & x_2 & \leq & -3 \\
& 2\,x_1 & + & x_2 & \leq & 4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



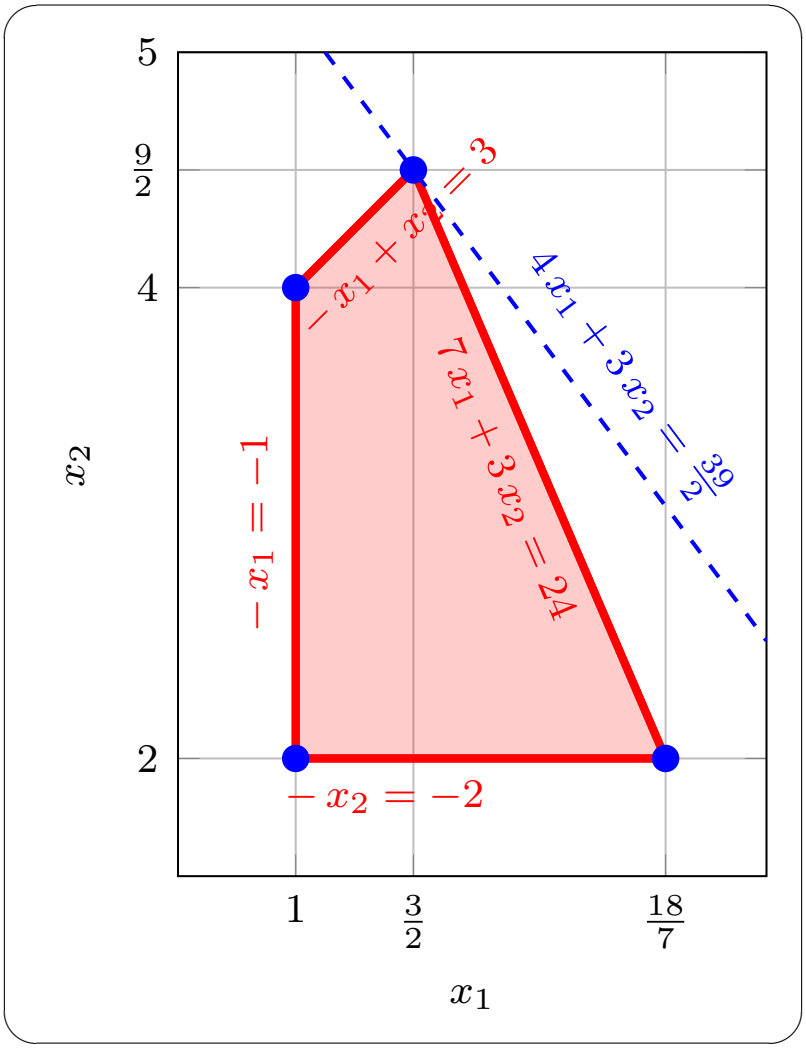
3.3 LP Problem *P23*

$$
\begin{array}{rrcrcr}
\max & x_1 & + & 4\,x_2 & & \\
\text{subject to} & 3\,x_1 & + & 8\,x_2 & \leq & 24 \\
 & x_1 & & & \geq & 1 \\
 & & & x_2 & \geq & 1 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



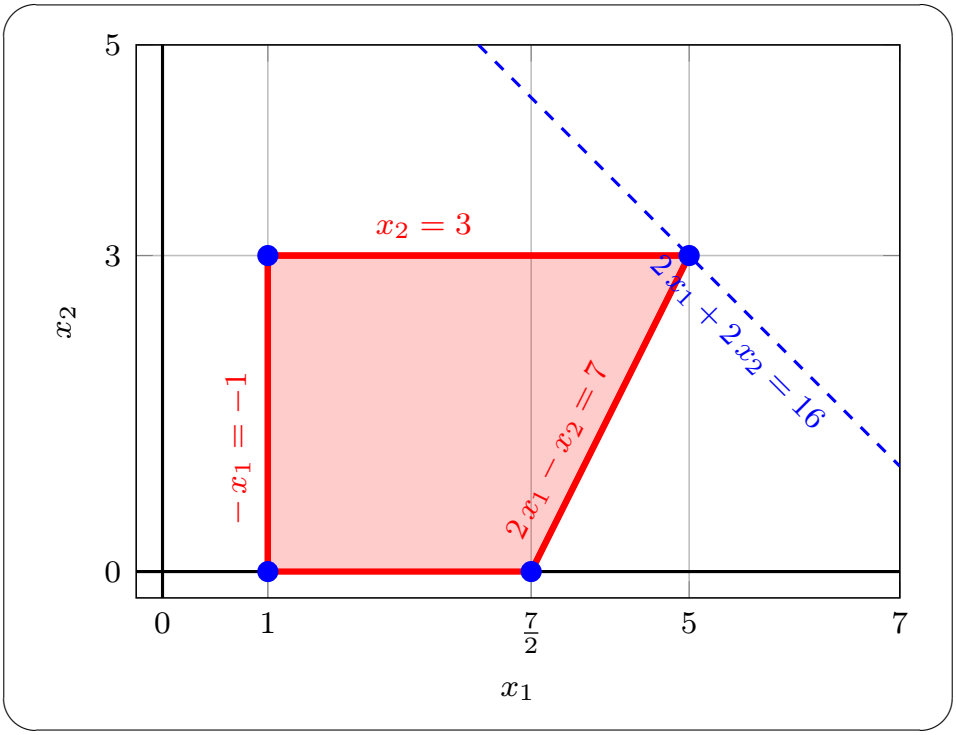
3.4 LP Problem *P24*

$$
\begin{array}{rrcrcr}
\max & 4\,x_1 & + & 3\,x_2 & & \\
\text{subject to} & 7\,x_1 & + & 3\,x_2 & \leq & 24 \\
& -\,x_1 & + & x_2 & \leq & 3 \\
& x_1 & & & \geq & 1 \\
& & & x_2 & \geq & 2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



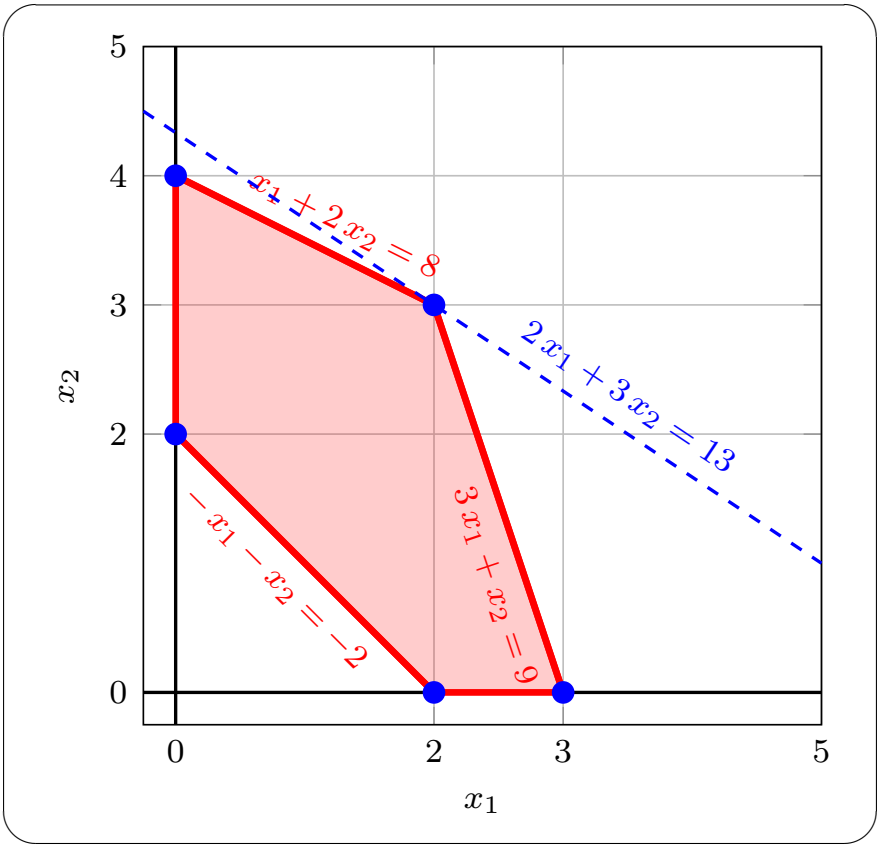
3.5 LP Problem *P25*

$$
\begin{array}{rrcrcl}
\max & 2\,x_1 & + & 2\,x_2 & & \\
\text{subject to} & x_1 & & & \geq & 1 \\
 & & & x_2 & \leq & 3 \\
 & 2\,x_1 & - & x_2 & \leq & 7 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



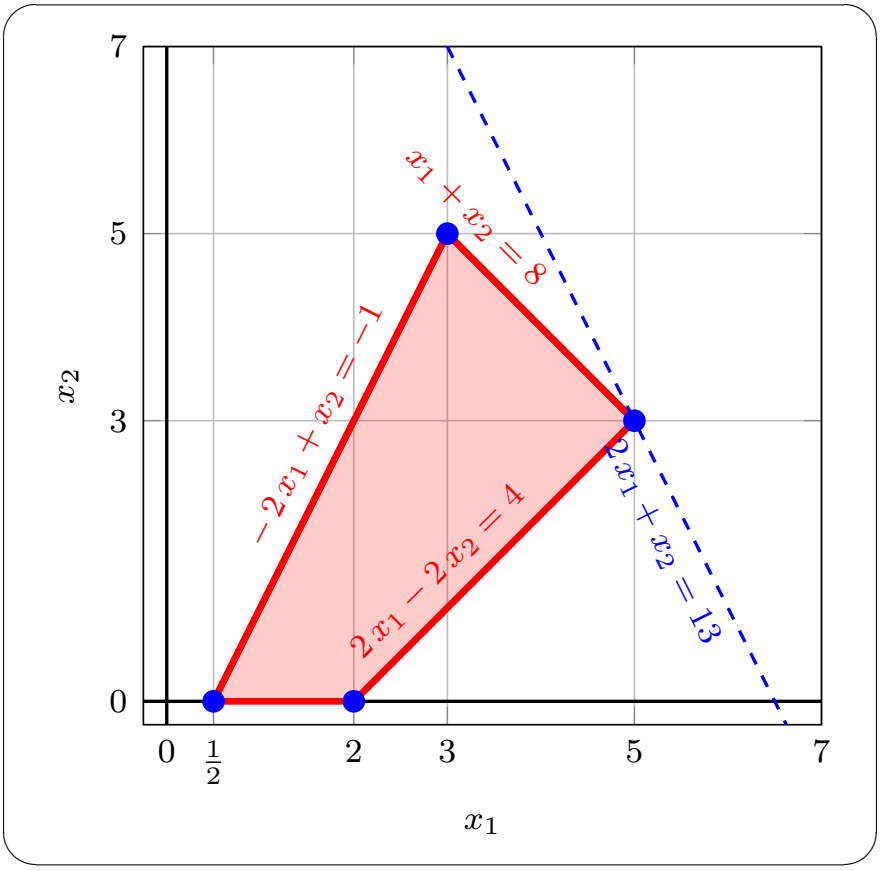
3.6 LP Problem *P26*

$$
\begin{array}{rrcrcr}
\max & 2\,x_1 & + & 3\,x_2 & & \\
\text{subject to} & -\,x_1 & - & x_2 & \leq & -2 \\
& x_1 & + & 2\,x_2 & \leq & 8 \\
& 3\,x_1 & + & x_2 & \leq & 9 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



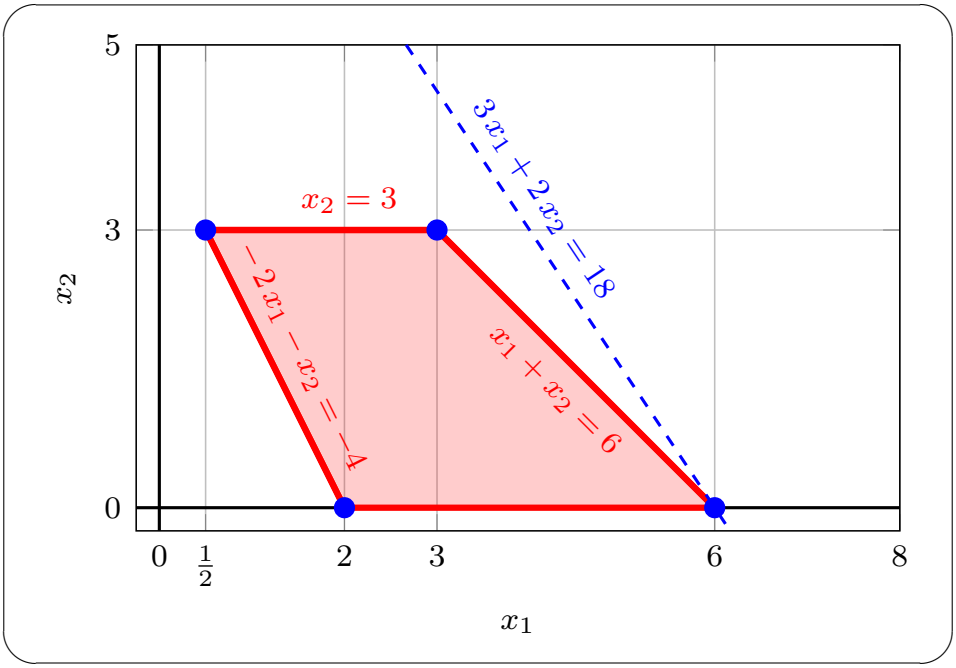
3.7 LP Problem *P27*

$$
\begin{array}{rrcrcr}
\max & 2\,x_1 & + & x_2 & & \\
\text{subject to} & x_1 & + & x_2 & \leq & 8 \\
& 2\,x_1 & - & 2\,x_2 & \leq & 4 \\
& -2\,x_1 & + & x_2 & \leq & -1 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



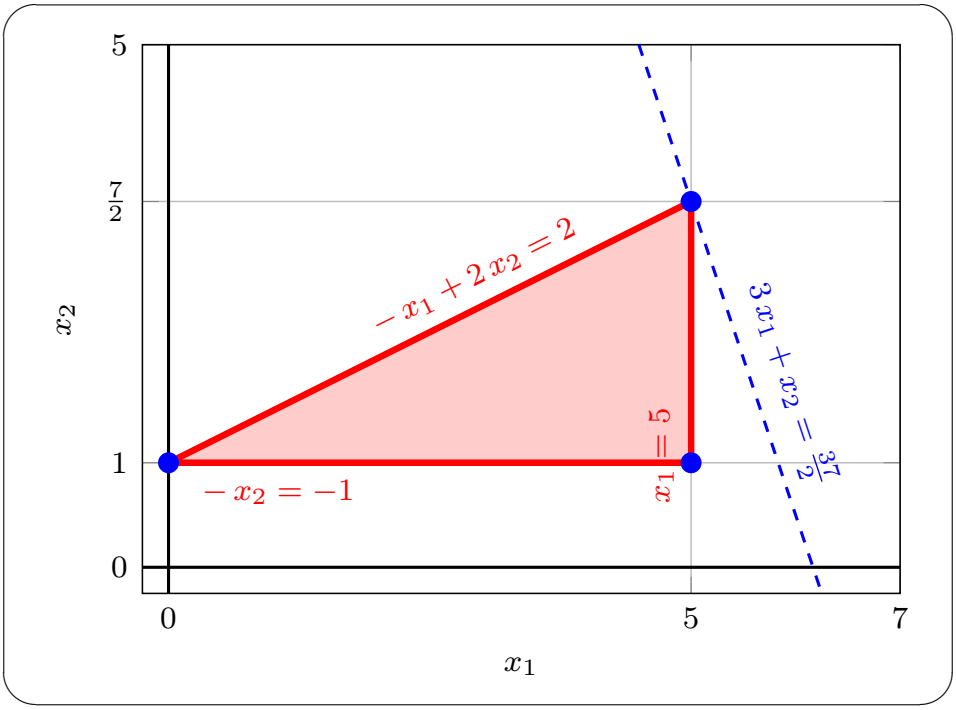
3.8 LP Problem *P28*

$$
\begin{array}{rrcrcr}
\max & 3\,x_1 & + & 2\,x_2 & & \\
\text{subject to} & x_1 & + & x_2 & \leq & 6 \\
& -2\,x_1 & - & x_2 & \leq & -4 \\
& & & x_2 & \leq & 3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



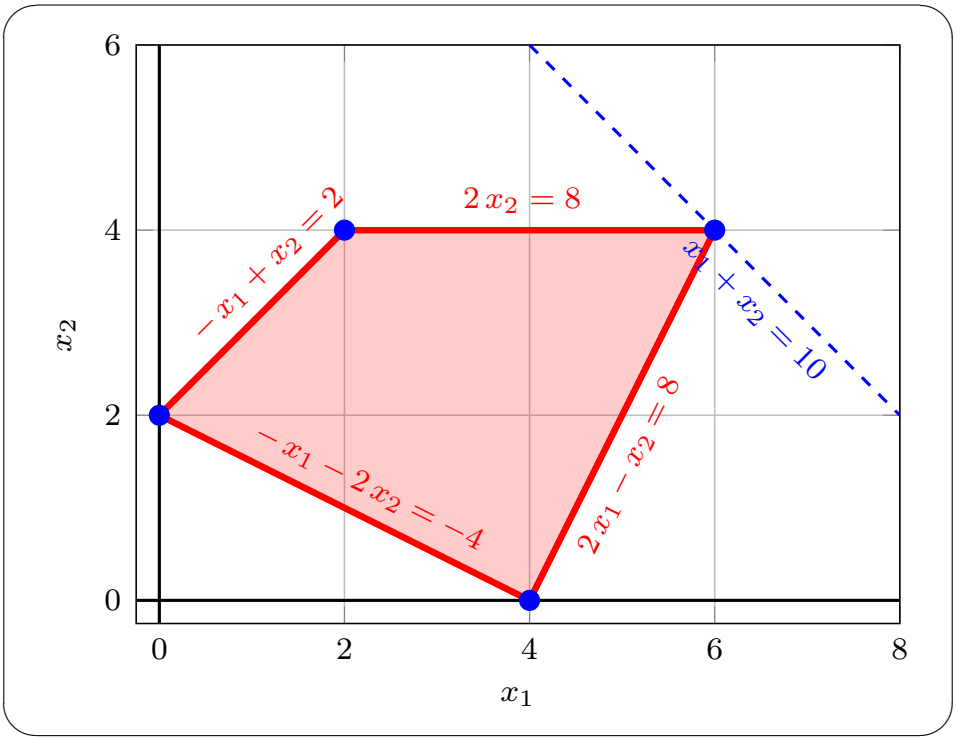
3.9 LP Problem *P29*

$$
\begin{array}{rrcrcl}
\max & 3\,x_1 & + & x_2 & & \\
\text{subject to} & -\,x_1 & + & 2\,x_2 & \leq & 2 \\
 & & & x_2 & \geq & 1 \\
 & x_1 & & & \leq & 5 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



3.10 LP Problem *P*30

$$
\begin{array}{rrrrrr}
\max & x_1 & + & x_2 & & \\
\text{subject to} & -x_1 & + & x_2 & \leq & 2 \\
& -x_1 & - & 2\,x_2 & \leq & -4 \\
& 2\,x_1 & - & x_2 & \leq & 8 \\
& & & 2\,x_2 & \leq & 8 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$

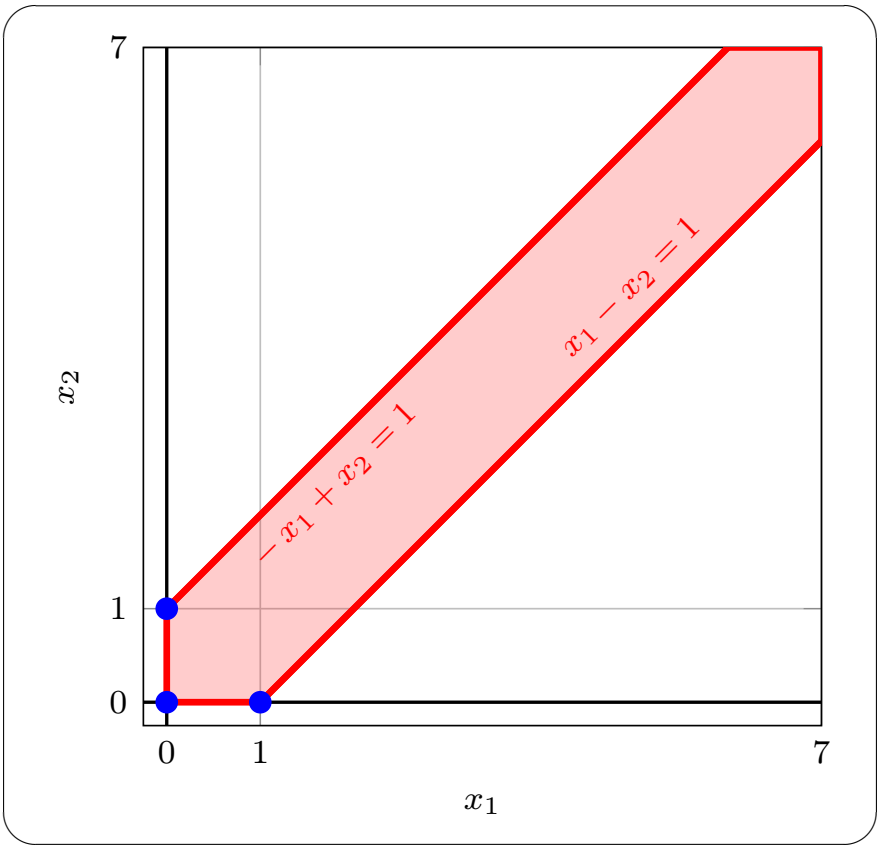


Chapter 4

LP problems with two variables – Type D

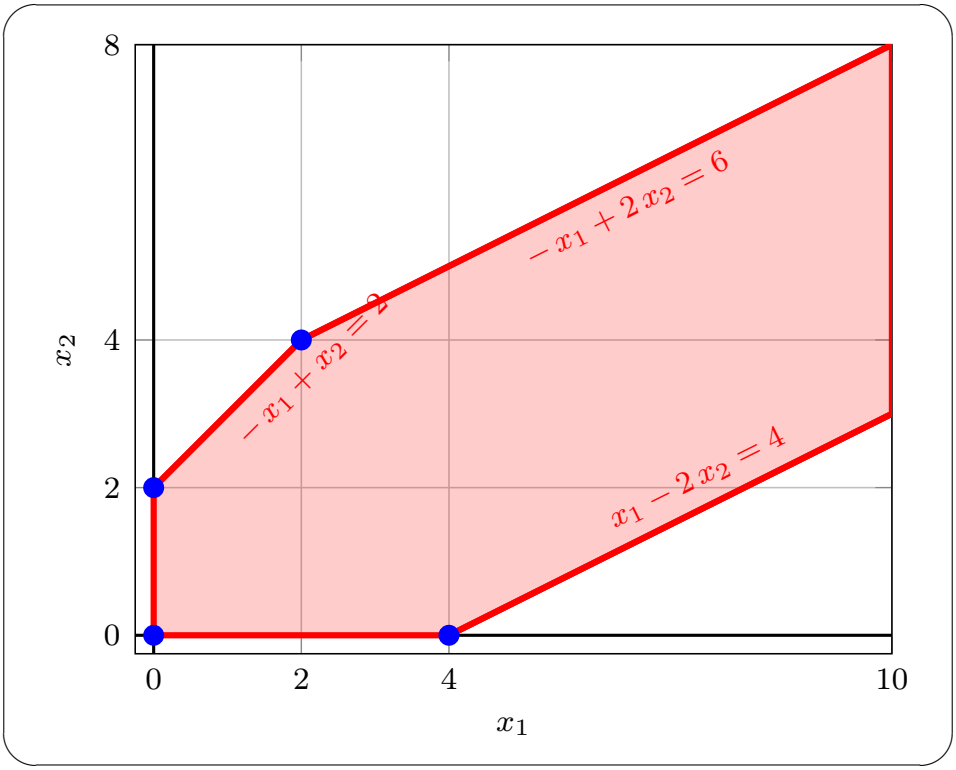
4.1 LP Problem *P31*

$$
\begin{array}{rrcrcl}
\max & x_1 & + & x_2 & & \\
\text{subject to} & -x_1 & + & x_2 & \leq & 1 \\
& x_1 & - & x_2 & \leq & 1 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



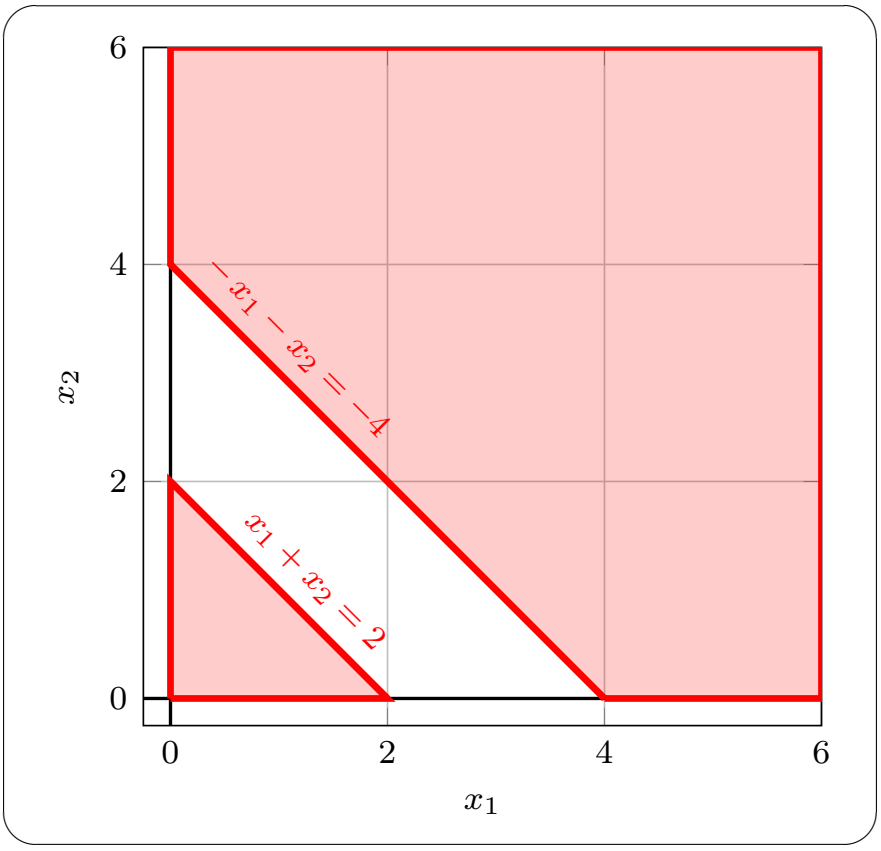
4.2 LP Problem *P32*

$$
\begin{array}{rrcrcl}
\max & 2\,x_1 & + & x_2 & & \\
\text{subject to} & -\,x_1 & + & x_2 & \leq & 2 \\
& -\,x_1 & + & 2\,x_2 & \leq & 6 \\
& x_1 & - & 2\,x_2 & \leq & 4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



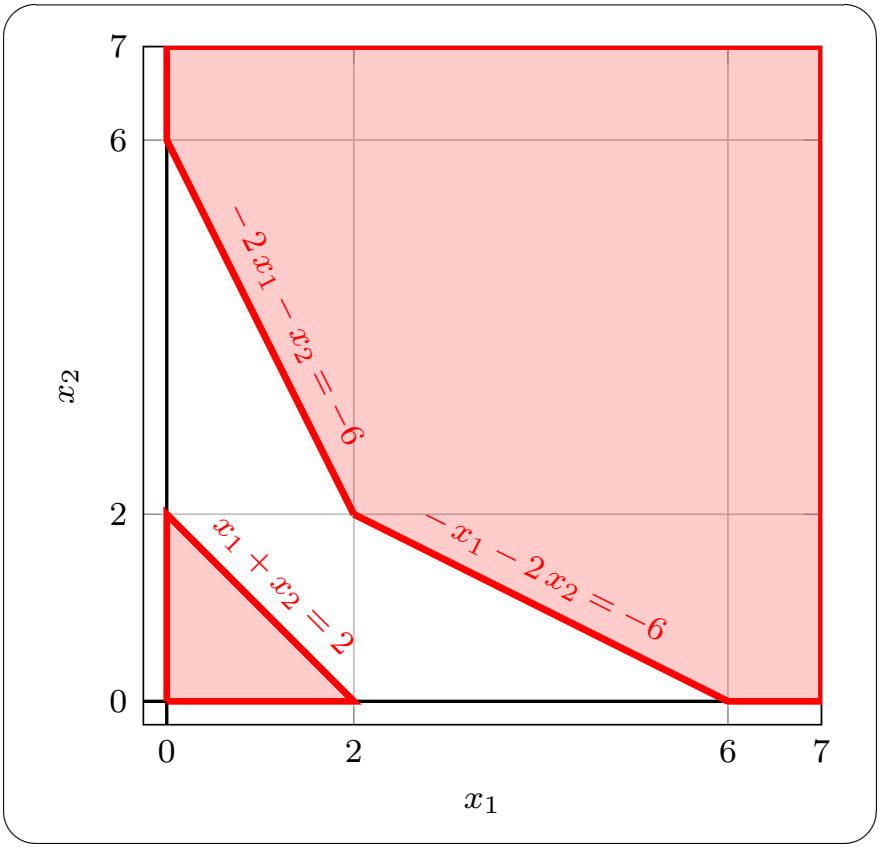
4.3 LP Problem *P33*

$$
\begin{array}{rrcrcr}
\max & x_1 & + & x_2 & & \\
\text{subject to} & x_1 & + & x_2 & \leq & 2 \\
& -x_1 & - & x_2 & \leq & -4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



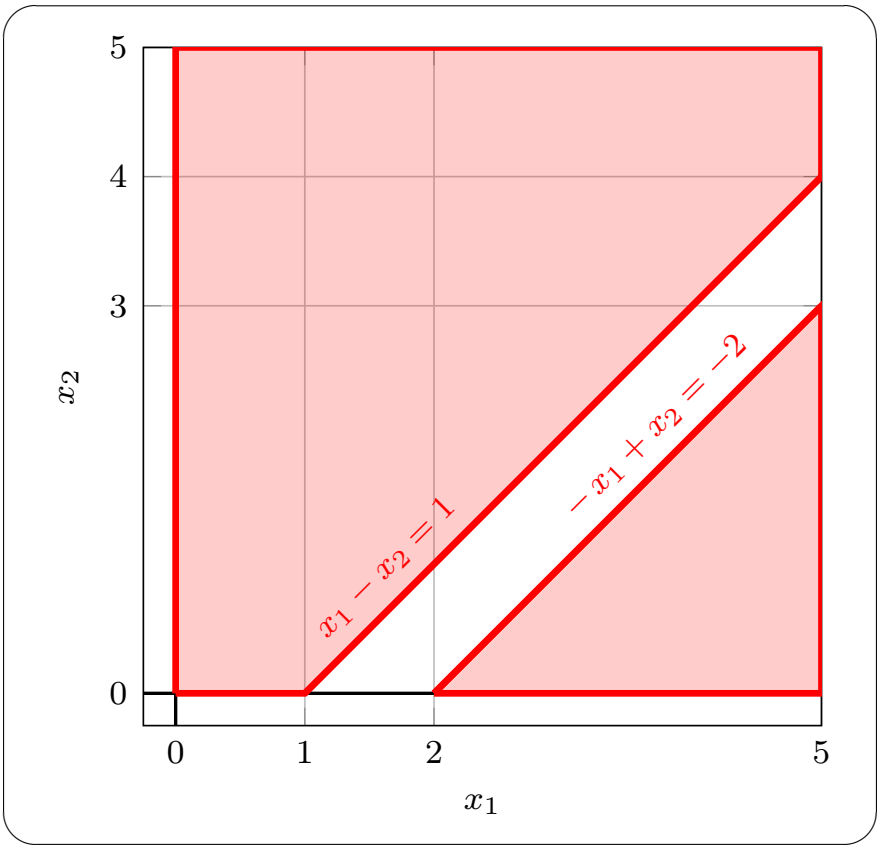
4.4 LP Problem *P34*

$$
\begin{array}{rrrrrr}
\max & -2\,x_1 & - & x_2 & & \\
\text{subject to} & -\,x_1 & - & 2\,x_2 & \leq & -6 \\
& -2\,x_1 & - & x_2 & \leq & -6 \\
& x_1 & + & x_2 & \leq & 2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



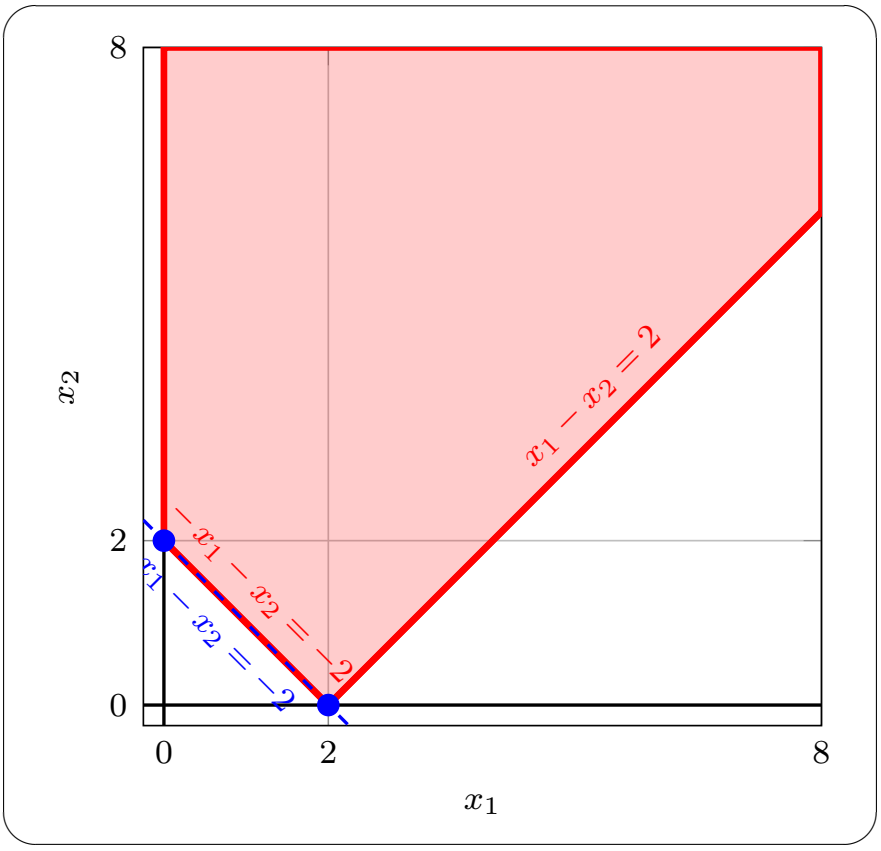
4.5 LP Problem *P35*

$$
\begin{array}{rrcrcr}
\max & x_1 & + & x_2 & & \\
\text{subject to} & x_1 & - & x_2 & \leq & 1 \\
& -x_1 & + & x_2 & \leq & -2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



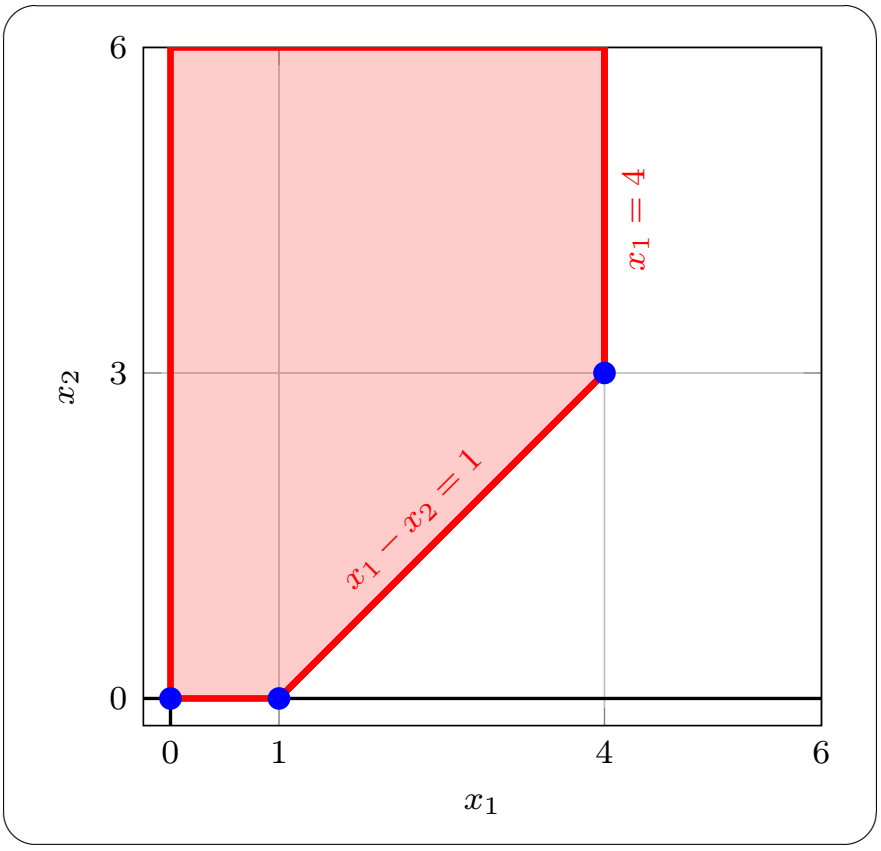
4.6 LP Problem *P36*

$$
\begin{array}{rrrrrr}
\max & -x_1 & - & x_2 & & \\
\text{subject to} & -x_1 & - & x_2 & \leq & -2 \\
& x_1 & - & x_2 & \leq & 2 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



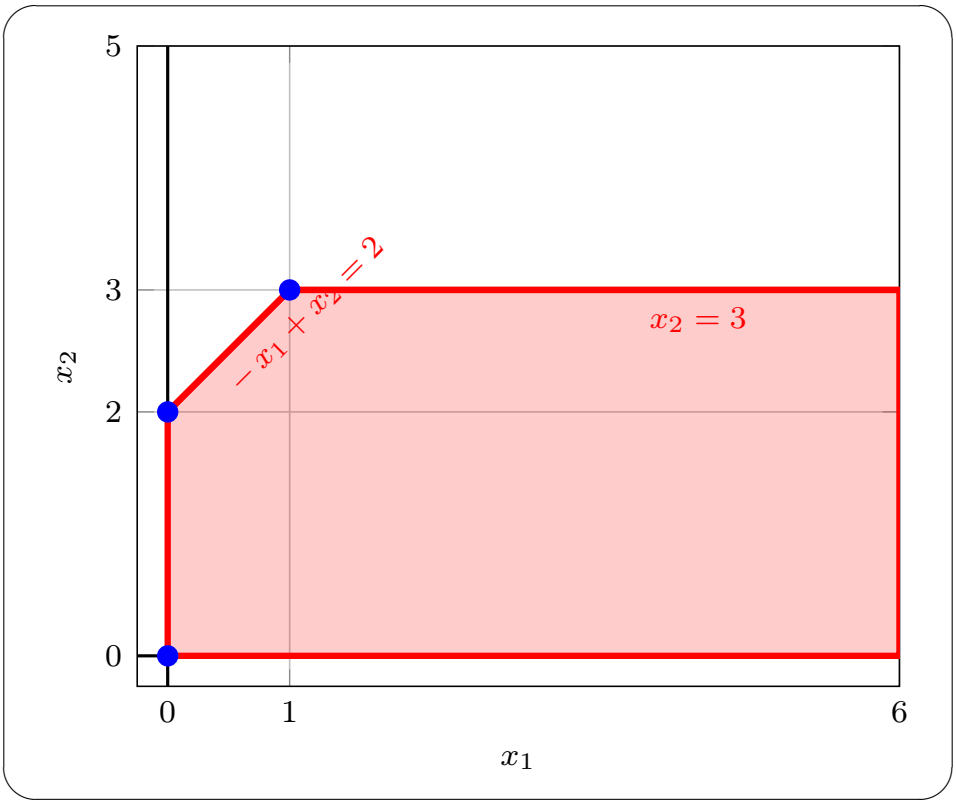
4.7 LP Problem *P37*

$$
\begin{array}{rrrrrr}
\max & x_1 & + & 2\,x_2 & & \\
\text{subject to} & x_1 & - & x_2 & \leq & 1 \\
& x_1 & & & \leq & 4 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



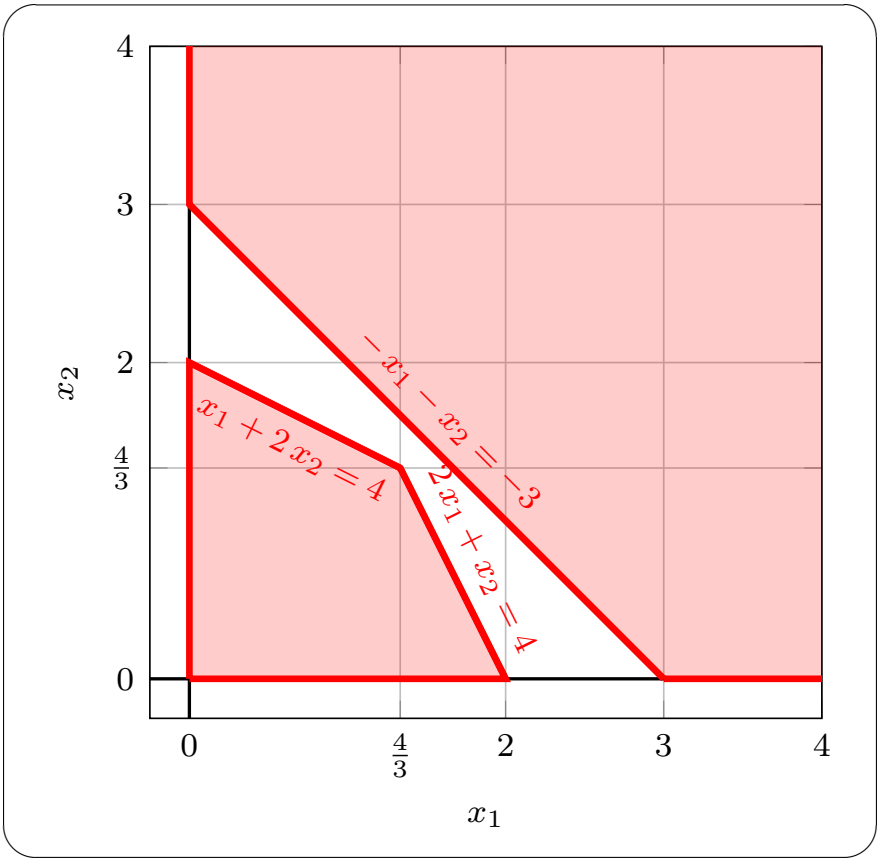
4.8 LP Problem *P38*

$$
\begin{array}{rrcrcl}
\max & 2\,x_1 & + & x_2 & & \\
\text{subject to} & -\,x_1 & + & x_2 & \leq & 2 \\
 & & & x_2 & \leq & 3 \\
 & x_1, & & x_2 & \geq & 0
\end{array}
$$



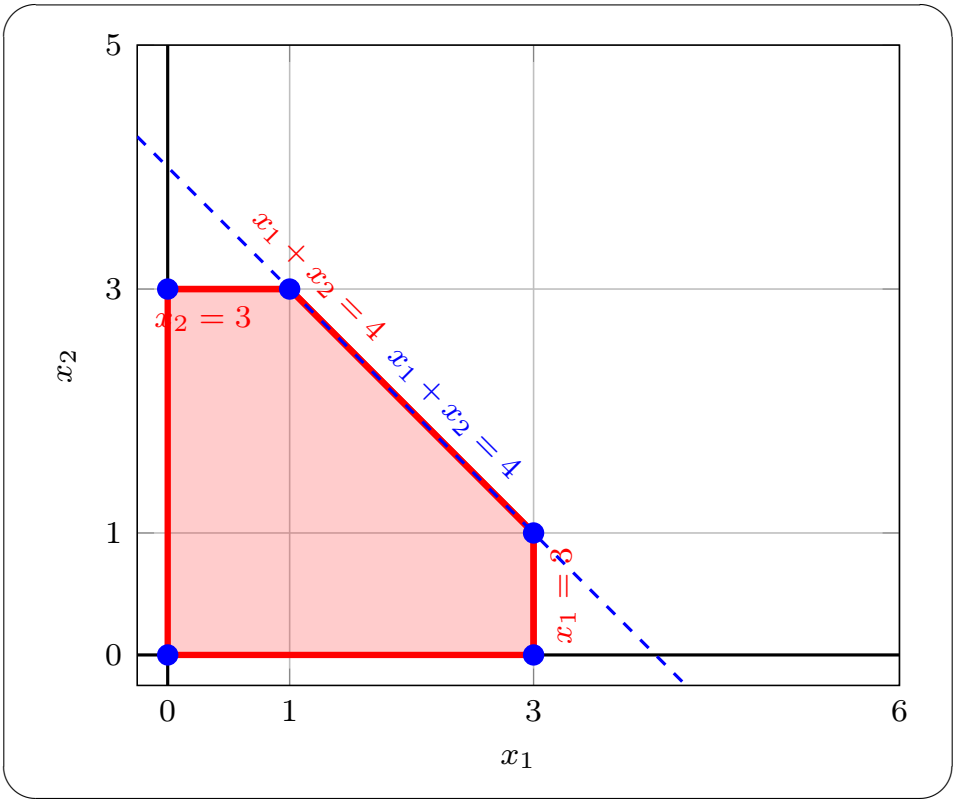
4.9 LP Problem *P39*

$$
\begin{array}{rrcrcr}
\max & x_1 & + & x_2 & & \\
\text{subject to} & 2\,x_1 & + & x_2 & \leq & 4 \\
& x_1 & + & 2\,x_2 & \leq & 4 \\
& -\,x_1 & - & x_2 & \leq & -3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$



4.10 LP Problem *P*40

$$
\begin{array}{rrcrcl}
\max & x_1 & + & x_2 & & \\
\text{subject to} & x_1 & + & x_2 & \leq & 4 \\
& x_1 & & & \leq & 3 \\
& & & x_2 & \leq & 3 \\
& x_1, & & x_2 & \geq & 0
\end{array}
$$

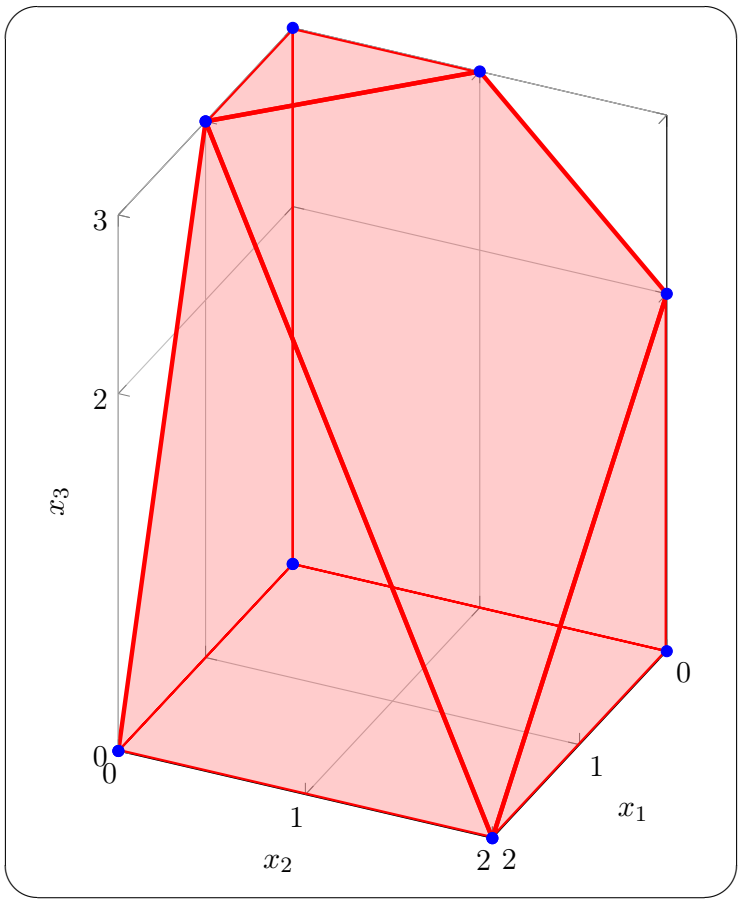


Chapter 5

LP problems with three variables – Type A

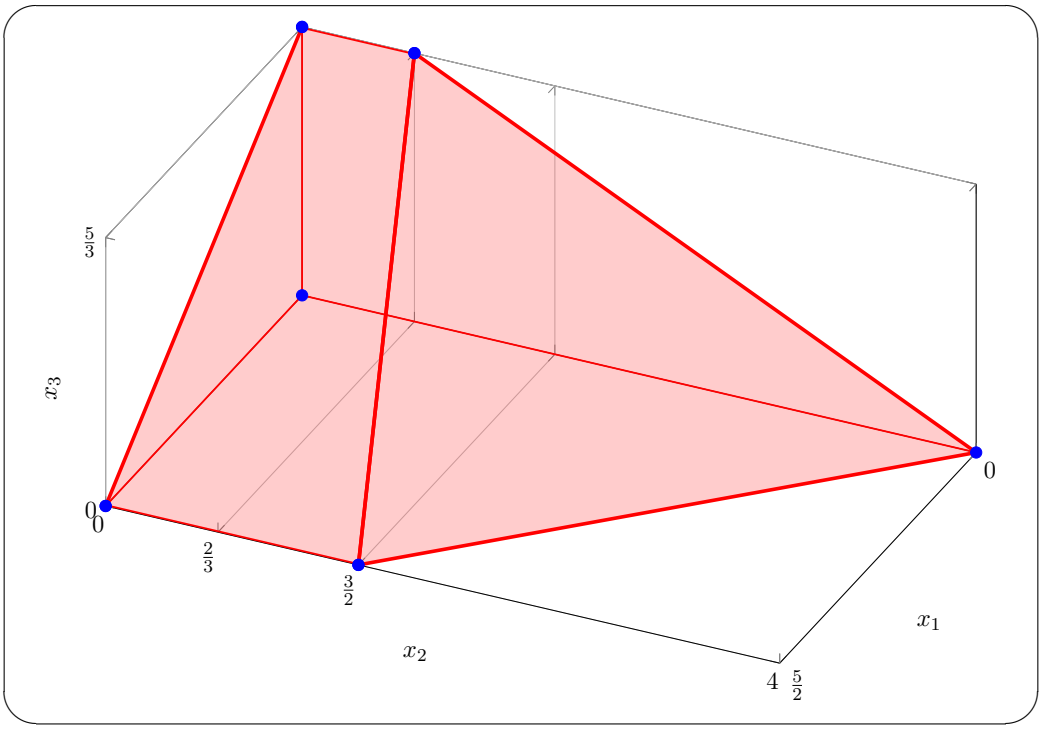
5.1 LP Problem *P41*

$$
\begin{array}{rrcrcrcl}
\max & 2\,x_1 & + & x_2 & + & 3\,x_3 & & \\
\text{subject to} & x_1 & + & x_2 & + & x_3 & \leq & 4 \\
 & & & x_2 & & & \leq & 2 \\
 & & & & & x_3 & \leq & 3 \\
 & 3\,x_1 & & & + & x_3 & \leq & 6 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



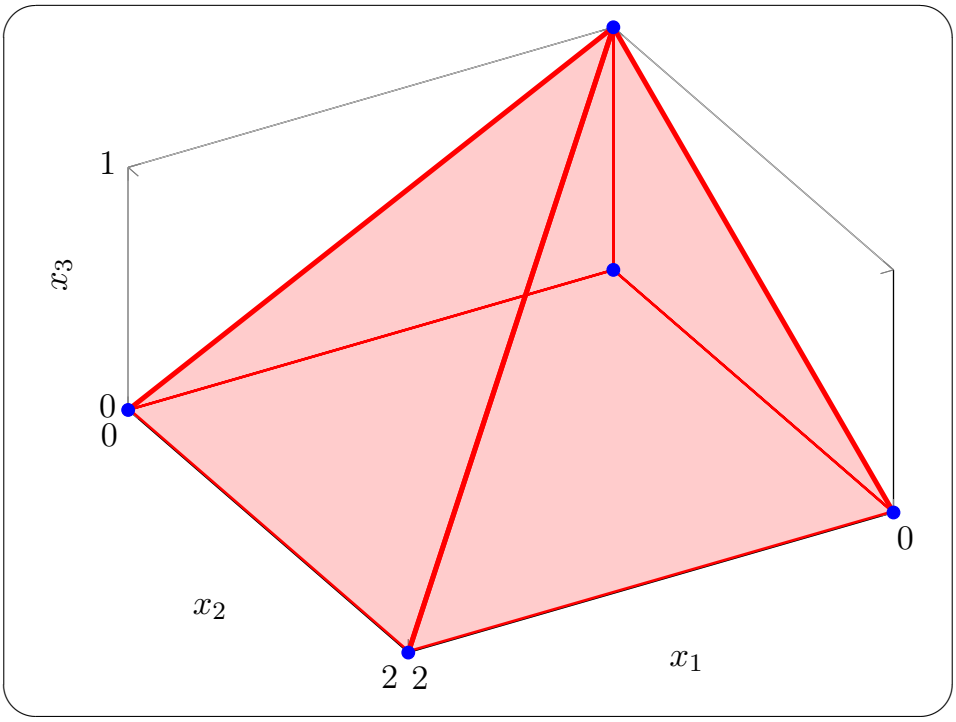
5.2 LP Problem *P42*

$$
\begin{array}{rrcrcrcrcl}
\max & 3 & x_1 & + & 2 & x_2 & + & 4 & x_3 & \\
\text{subject to} & & x_1 & + & & x_2 & + & 2 & x_3 & \leq & 4 \\
& 2 & x_1 & & & & + & 3 & x_3 & \leq & 5 \\
& 2 & x_1 & + & & x_2 & + & 3 & x_3 & \leq & 7 \\
& & x_1, & & & x_2, & & & x_3 & \geq & 0
\end{array}
$$



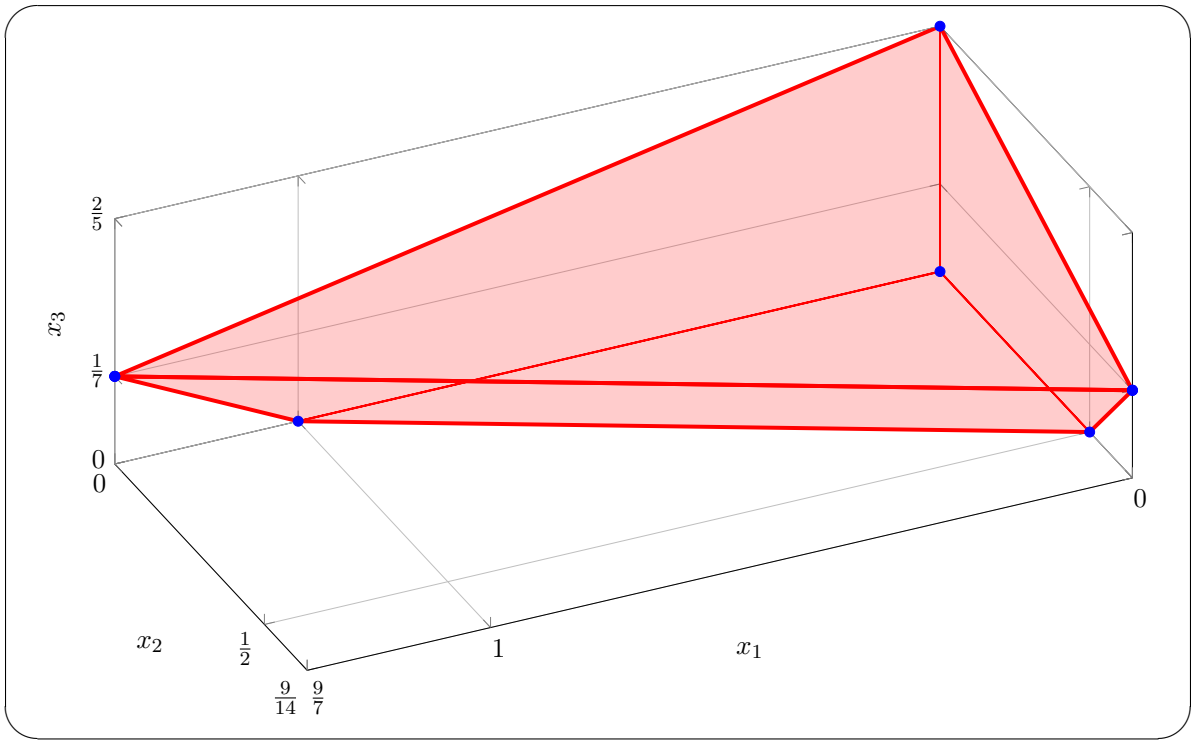
5.3 LP Problem *P43*

$$
\begin{array}{rrcrcrcrcl}
\max & x_1 & + & 2 & x_2 & + & 7 & x_3 & & \\
\text{subject to} & x_1 & & & & + & 2 & x_3 & \leq & 2 \\
& & & & x_2 & + & 2 & x_3 & \leq & 2 \\
& x_1, & & & x_2, & & & x_3 & \geq & 0
\end{array}
$$



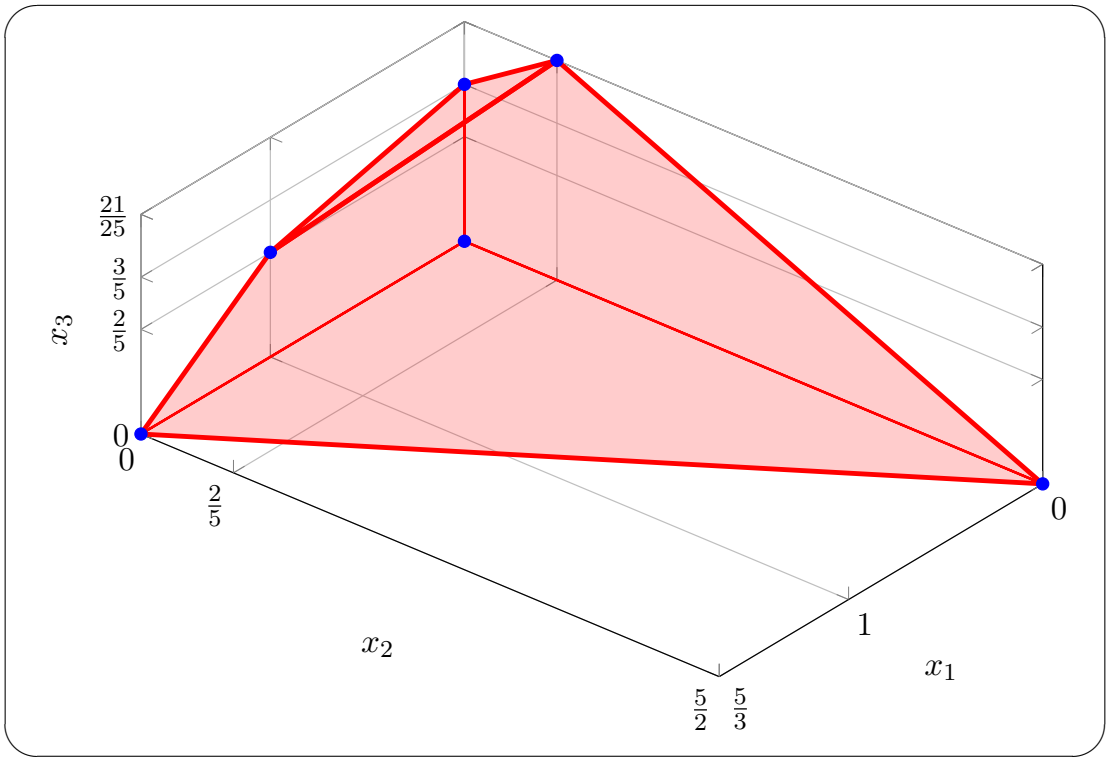
5.4 LP Problem *P44*

$$
\begin{array}{rrcrcrcrcl}
\max & x_1 & + & 2 & x_2 & + & 3 & x_3 & & \\
\text{subject to} & x_1 & + & 2 & x_2 & - & 2 & x_3 & \leq & 1 \\
2 & x_1 & - & & x_2 & - & 3 & x_3 & \leq & 4 \\
& x_1 & + & 2 & x_2 & + & 5 & x_3 & \leq & 2 \\
& x_1, & & & x_2, & & & x_3 & \geq & 0
\end{array}
$$



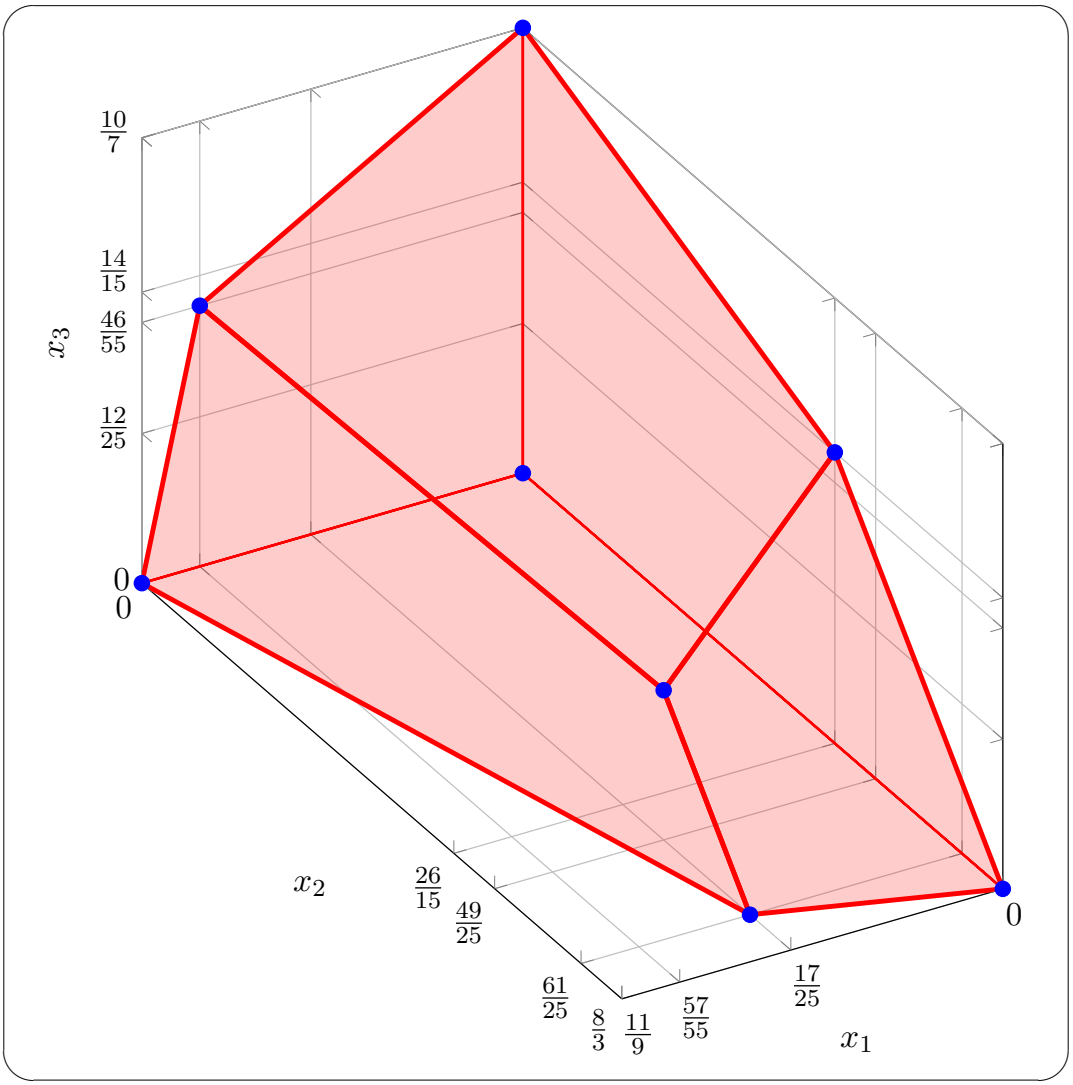
5.5 LP Problem *P45*

$$
\begin{array}{rrcrcrcl}
\max & 4\,x_1 & - & 13\,x_2 & + & 7\,x_3 & & \\
\text{subject to} & 3\,x_1 & + & 2\,x_2 & + & 5\,x_3 & \leq & 5 \\
& x_1 & - & 3\,x_2 & + & 5\,x_3 & \leq & 3 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



5.6 LP Problem *P46*

$$
\begin{array}{rrcrcrcl}
\max & 5\ x_1 & + & 6\ x_2 & + & 10\ x_3 & & \\
\text{subject to} & 9\ x_1 & + & 2\ x_2 & + & 2\ x_3 & \leq & 11 \\
& 2\ x_1 & + & 6\ x_2 & + & 6\ x_3 & \leq & 16 \\
& 4\ x_1 & + & 2\ x_2 & + & 7\ x_3 & \leq & 10 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

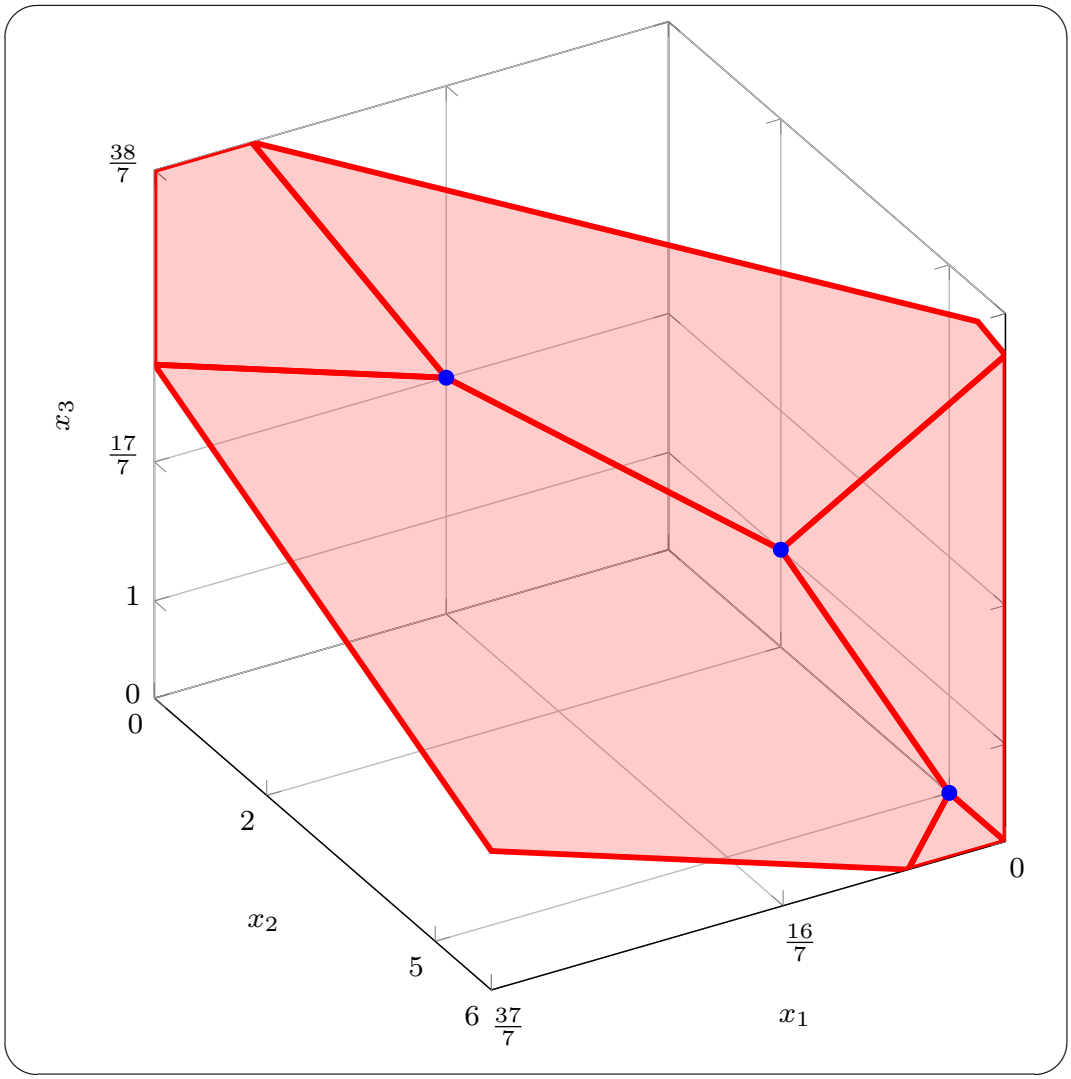


Chapter 6

LP problems with three variables – Type B

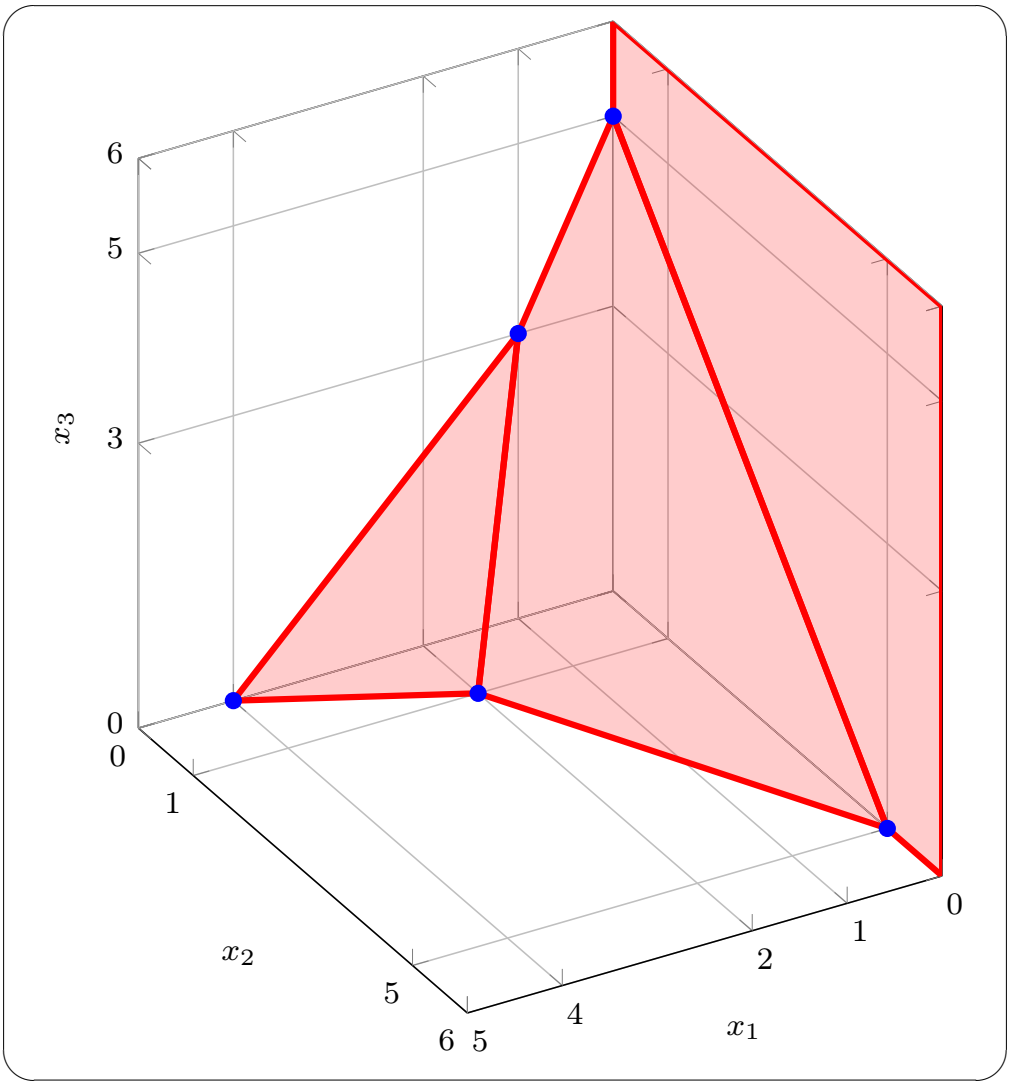
6.1 LP Problem $P51$

$$
\begin{array}{rrrrrrrrrrr}
\max & - & 2\,x_1 & - & x_2 & & & & & & \\
\text{subject to} & & x_1 & - & x_2 & - & 3 & x_3 & \leq & -5 \\
& - & 3 & x_1 & - & 2 & x_2 & + & 2 & x_3 & \leq & -2 \\
& & x_1, & & & & x_2, & & & x_3 & \geq & 0
\end{array}
$$



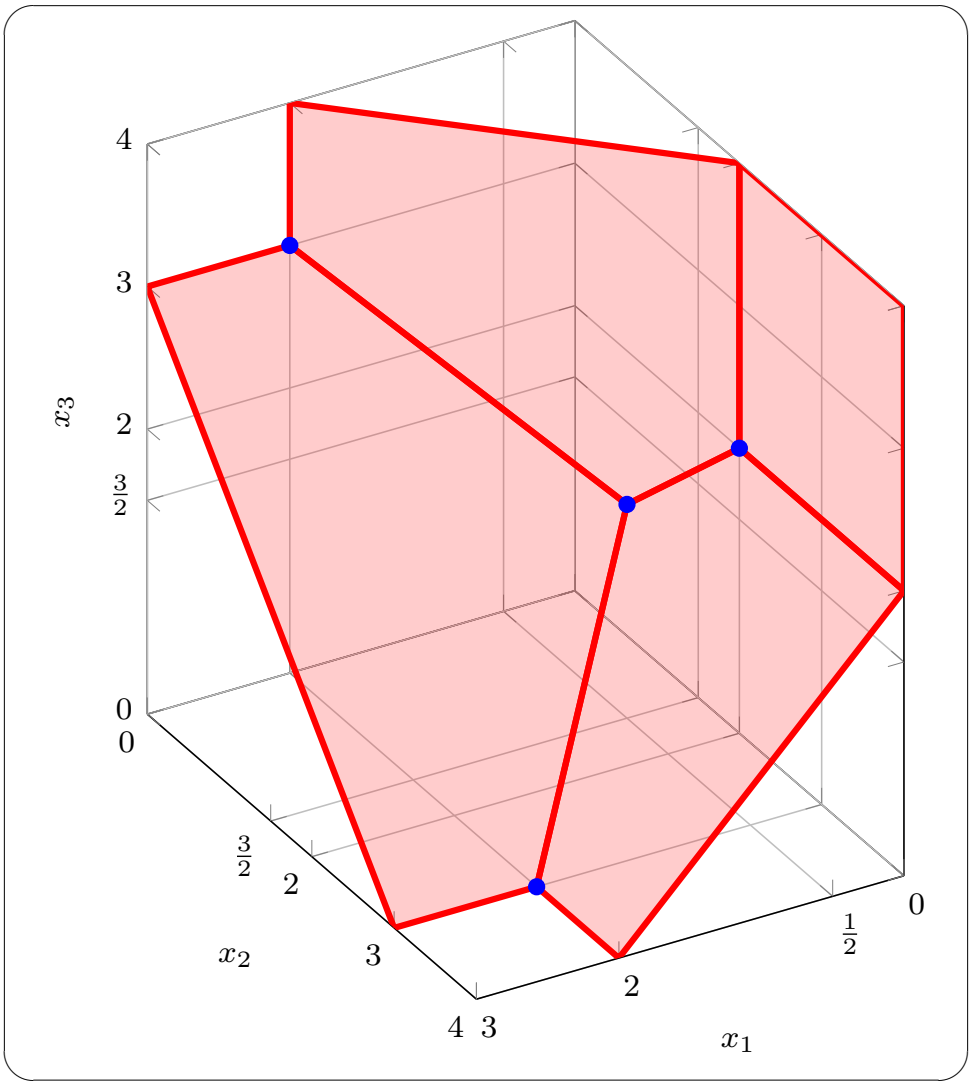
6.2 LP Problem *P52*

$$
\begin{array}{rrrrrrrrrr}
\max & - & x_1 & - & 2\,x_2 & - & x_3 & & \\
\text{subject to} & - & 2\,x_1 & - & x_2 & - & x_3 & \leq & -5 \\
& - & x_1 & - & 2\,x_2 & - & x_3 & \leq & -4 \\
& - & x_1 & - & x_2 & - & 2\,x_3 & \leq & -3 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



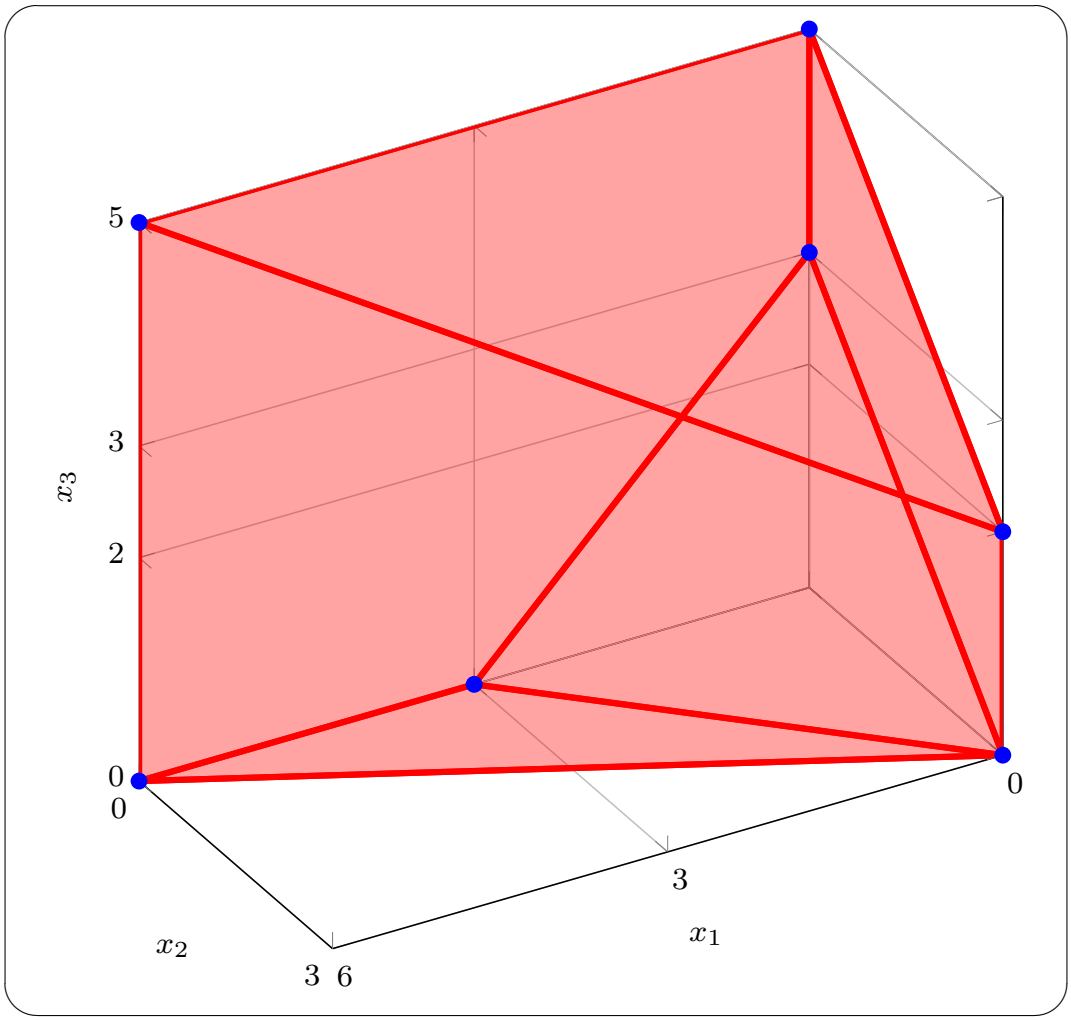
6.3 LP Problem *P53*

$$
\begin{array}{rrrrrrrrr}
\max & - & x_1 & - & 2\,x_2 & - & 4\,x_3 & & \\
\text{subject to} & - & x_1 & - & x_2 & & & \leq & -2 \\
 & & & - & x_2 & - & x_3 & \leq & -3 \\
 & - & x_1 & & & - & x_3 & \leq & -2 \\
 & & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



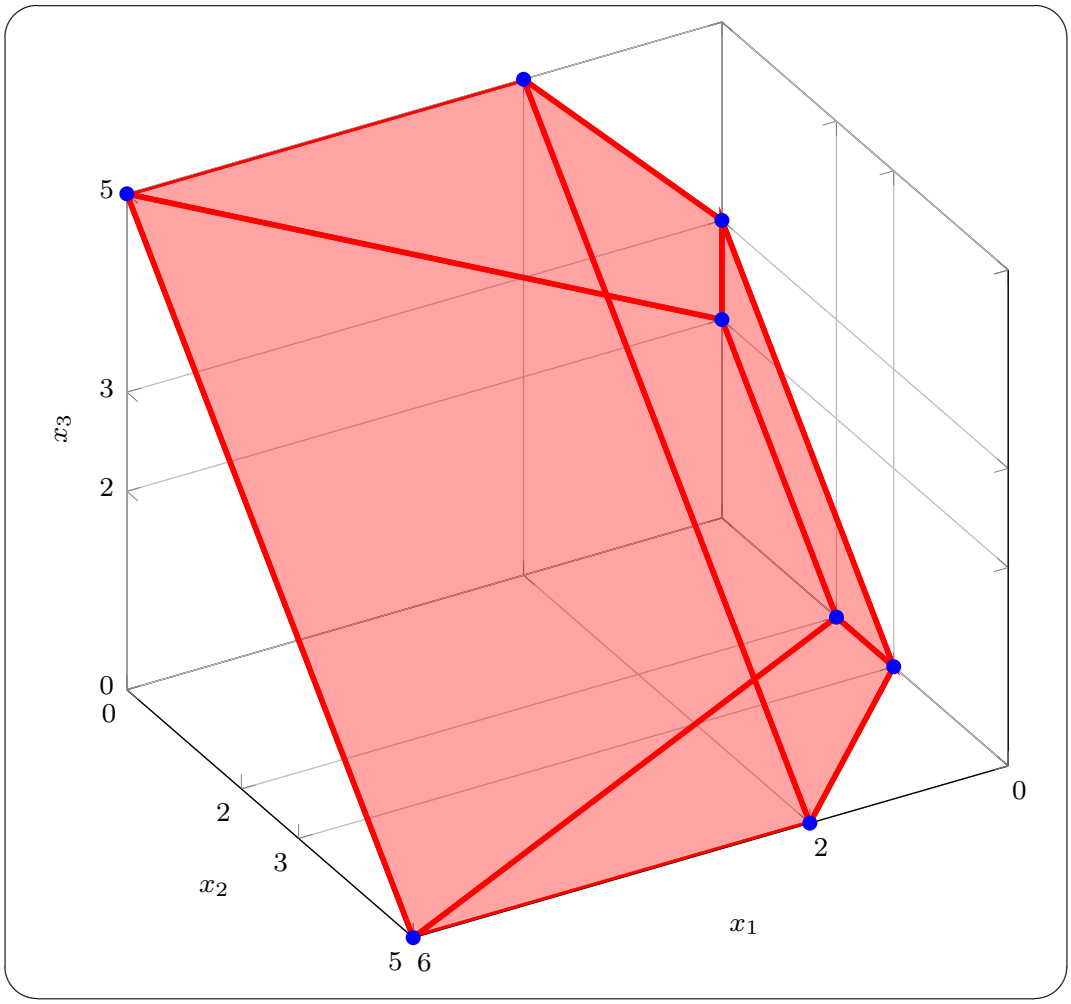
6.4 LP Problem *P54*

$$
\begin{array}{rrrrrrrrrr}
\max & - & x_1 & - & 2\,x_2 & - & 2\,x_3 & & & \\
\text{subject to} & & x_1 & + & 2\,x_2 & & & \leq & 6 \\
& - & x_1 & - & x_2 & - & x_3 & \leq & -3 \\
& & & & x_2 & + & x_3 & \leq & 5 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



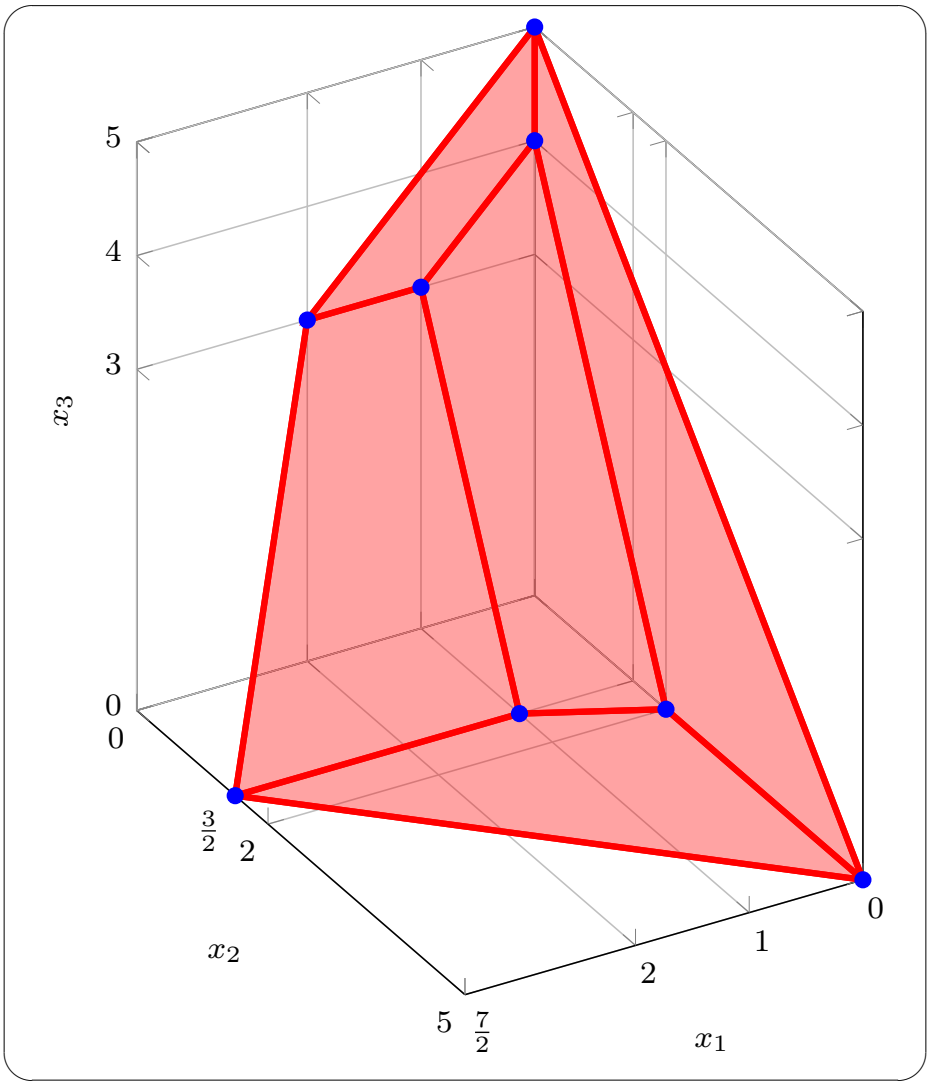
6.5 LP Problem *P55*

$$
\begin{array}{rrrrrrrrr}
\max & - & 2\,x_1 & - & 2\,x_2 & - & x_3 & & \\
\text{subject to} & & & & x_2 & + & x_3 & \leq & 5 \\
& - & x_1 & + & x_2 & + & x_3 & \leq & 3 \\
& & x_1 & - & 2\,x_2 & - & 2\,x_3 & \leq & -4 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



6.6 LP Problem *P56*

$$
\begin{array}{rrrrrrrrr}
\max & - & 2\,x_1 & - & 3\,x_2 & - & 3\,x_3 & & \\
\text{subject to} & - & x_1 & - & 2\,x_2 & - & x_3 & \leq & -4 \\
 & & & - & 2\,x_2 & - & x_3 & \leq & -3 \\
 & & x_1 & + & x_2 & + & x_3 & \leq & 5 \\
 & & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

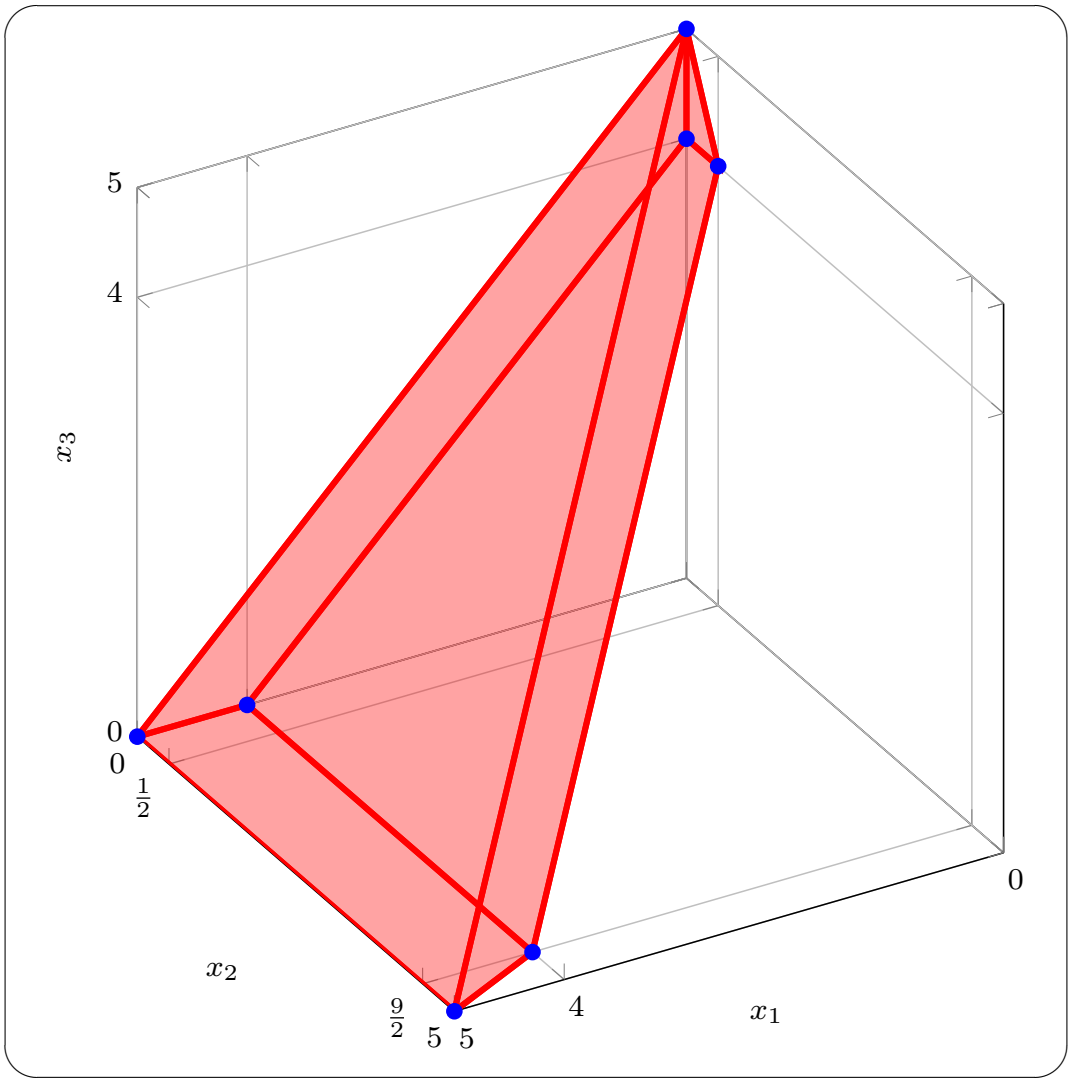


Chapter 7

LP problems with three variables – Type C

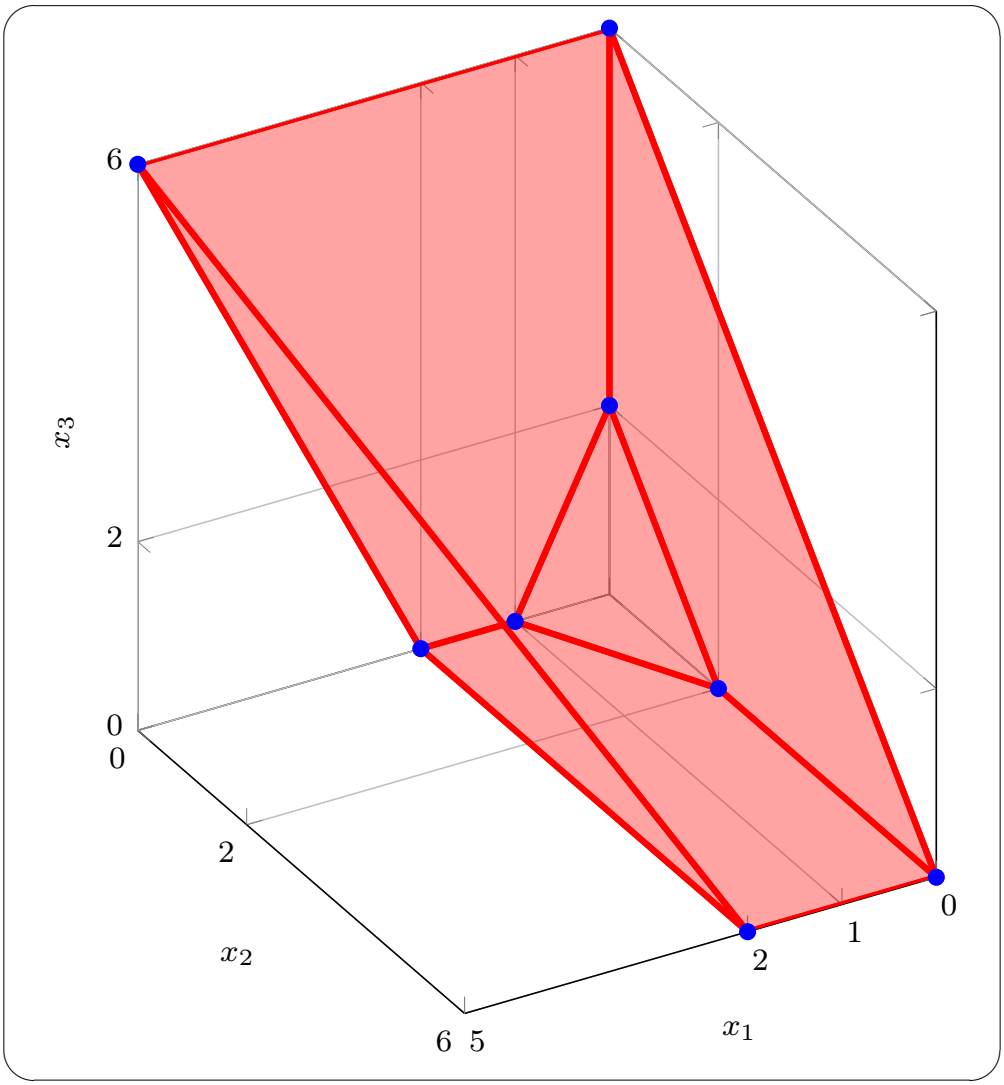
7.1 LP Problem *P61*

$$
\begin{array}{rrcrcrcr}
\max & 3\,x_1 & + & x_2 & + & 2\,x_3 & & \\
\text{subject to} & x_1 & & & + & x_3 & \leq & 5 \\
- & x_1 & & & - & x_3 & \leq & -4 \\
- & x_1 & + & 2\,x_2 & + & x_3 & \leq & 5 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



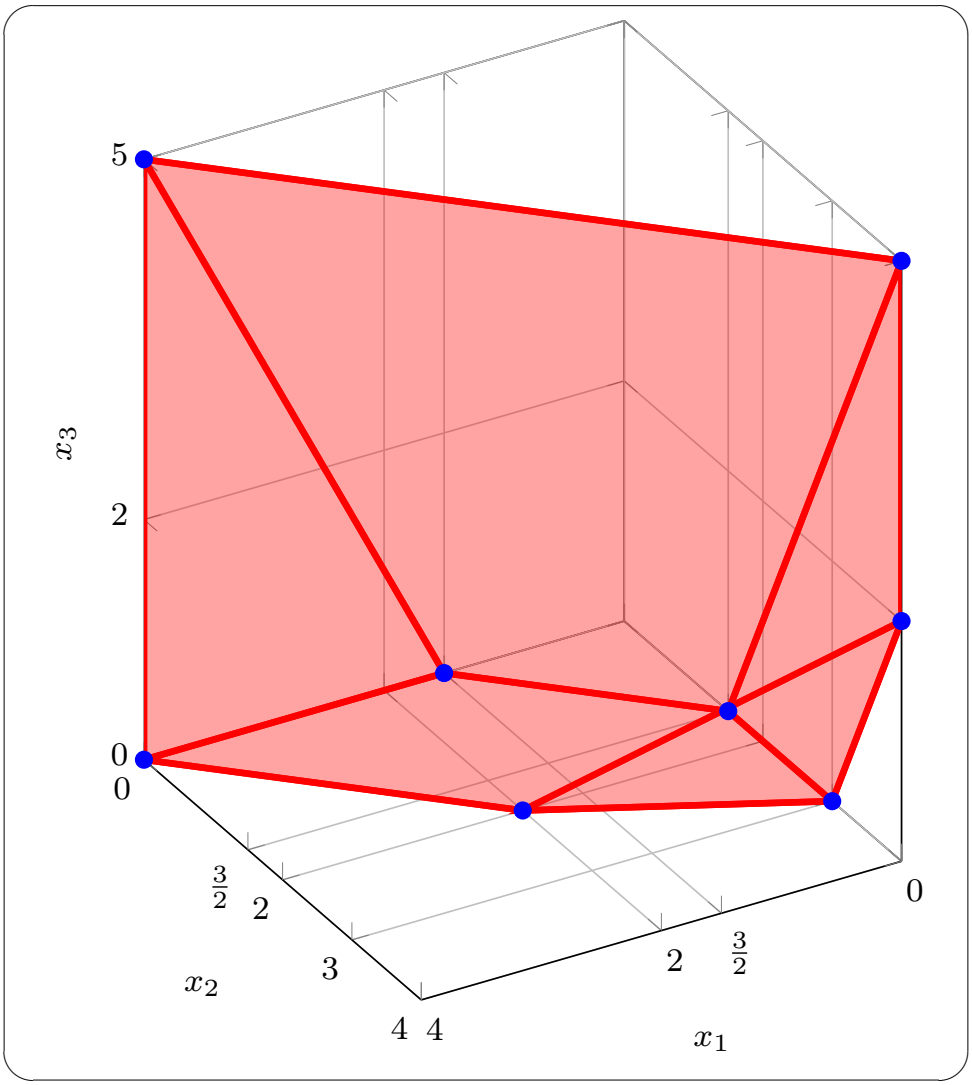
7.2 LP Problem *P62*

$$
\begin{array}{rrcrcrcr}
\max & 2\,x_1 & + & 3\,x_2 & + & x_3 & & \\
\text{subject to} & & & x_2 & + & x_3 & \leq & 6 \\
- & 2\,x_1 & - & x_2 & - & x_3 & \leq & -2 \\
& 2\,x_1 & & & - & x_3 & \leq & 4 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



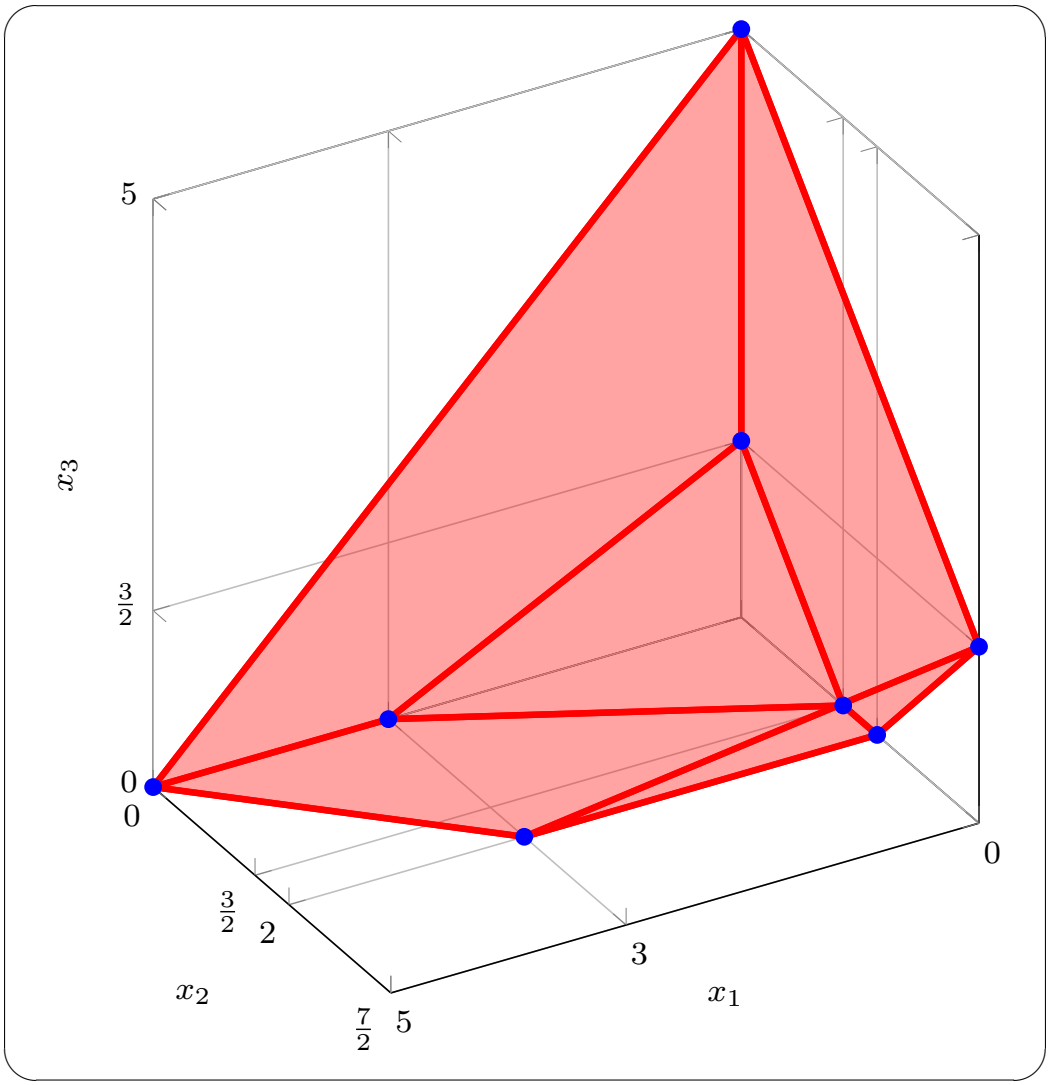
7.3 LP Problem *P63*

$$
\begin{array}{rrcrcrcr}
\max & x_1 & + & 2\,x_2 & + & 2\,x_3 & & \\
\text{subject to} & x_1 & + & x_2 & & & \leq & 4 \\
& x_1 & + & 2\,x_2 & - & x_3 & \leq & 6 \\
- & 2\,x_1 & - & 2\,x_2 & + & x_3 & \leq & -3 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



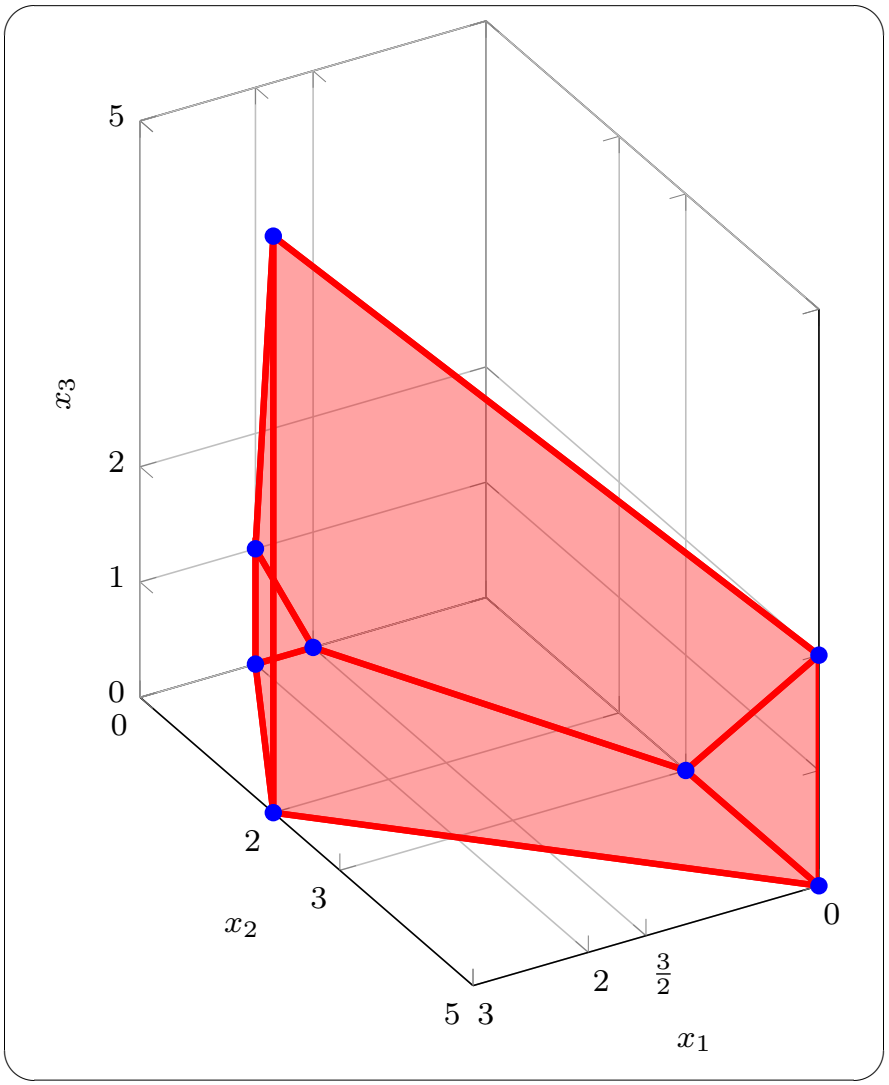
7.4 LP Problem *P64*

$$
\begin{array}{rrrrrrrrr}
\max & & 3\,x_1 & + & 2\,x_2 & + & x_3 & & \\
\text{subject to} & - & x_1 & - & 2\,x_2 & - & 2\,x_3 & \leq & -3 \\
 & & x_1 & + & x_2 & + & x_3 & \leq & 5 \\
 & & & & x_2 & - & x_3 & \leq & 2 \\
 & & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



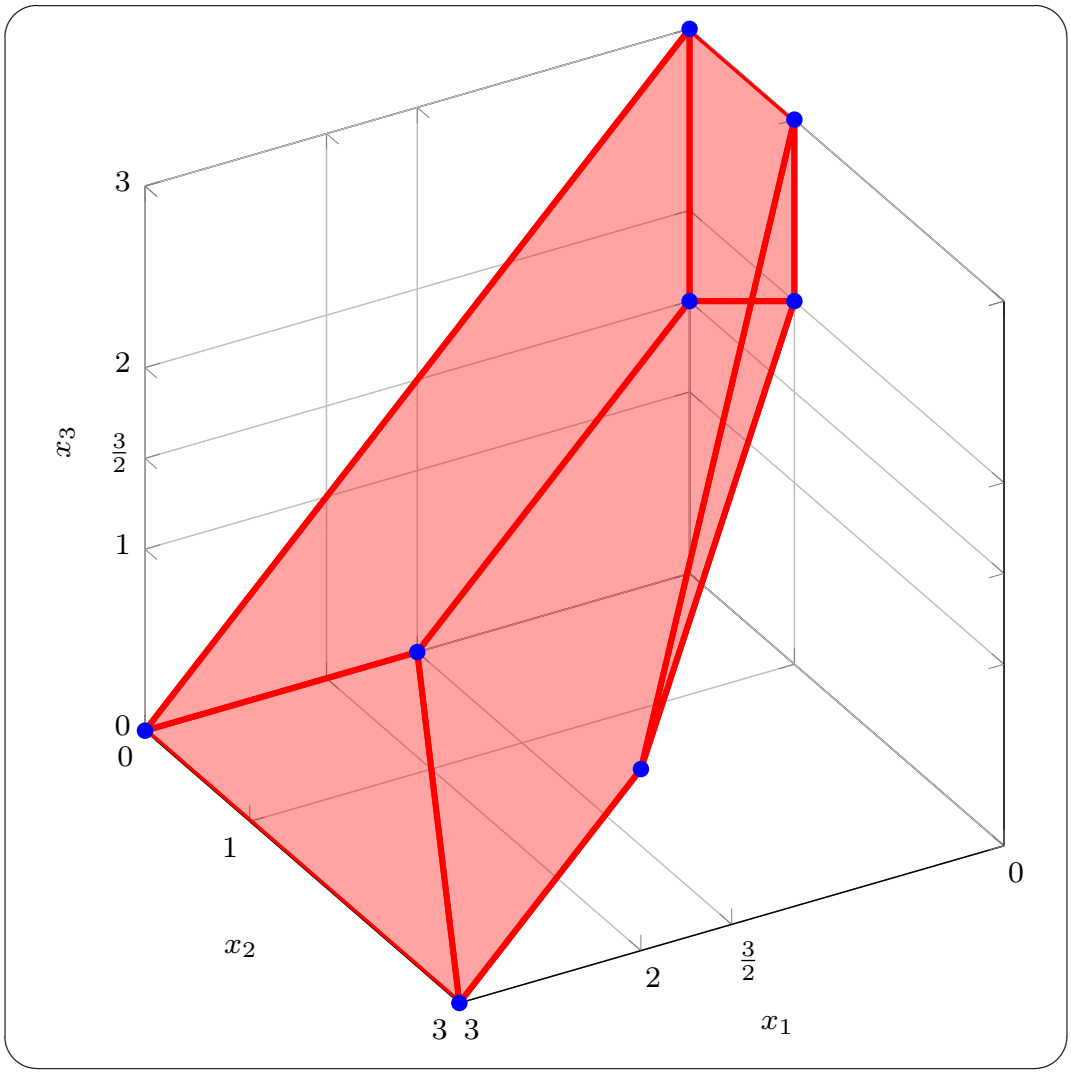
7.5 LP Problem *P65*

$$
\begin{array}{rrcrcrcr}
\max & 2\,x_1 & + & 3\,x_2 & + & 3\,x_3 & & \\
\text{subject to} & x_1 & + & x_2 & & & \leq & 5 \\
& 2\,x_1 & - & x_2 & & & \leq & 4 \\
- & 2\,x_1 & - & x_2 & + & x_3 & \leq & -3 \\
& x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



7.6 LP Problem *P66*

$$
\begin{array}{rrrrrrrrr}
\max & & 2\,x_1 & + & x_2 & + & x_3 & & \\
\text{subject to} & - & 2\,x_1 & + & x_2 & - & 2\,x_3 & \leq & -3 \\
& & x_1 & & & + & x_3 & \leq & 3 \\
& - & 2\,x_1 & + & 2\,x_2 & & & \leq & 2 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

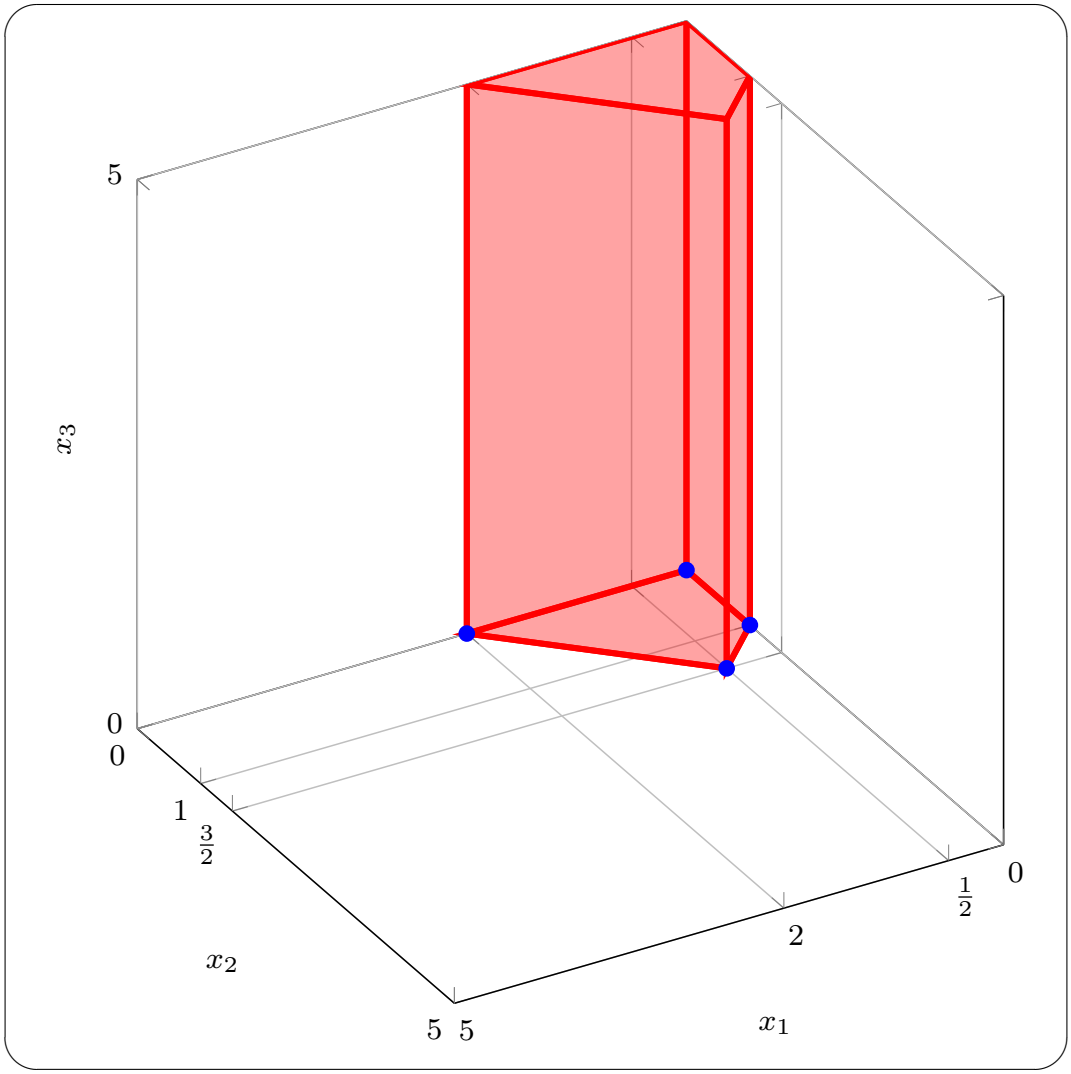


Chapter 8

LP problems with three variables – Type D

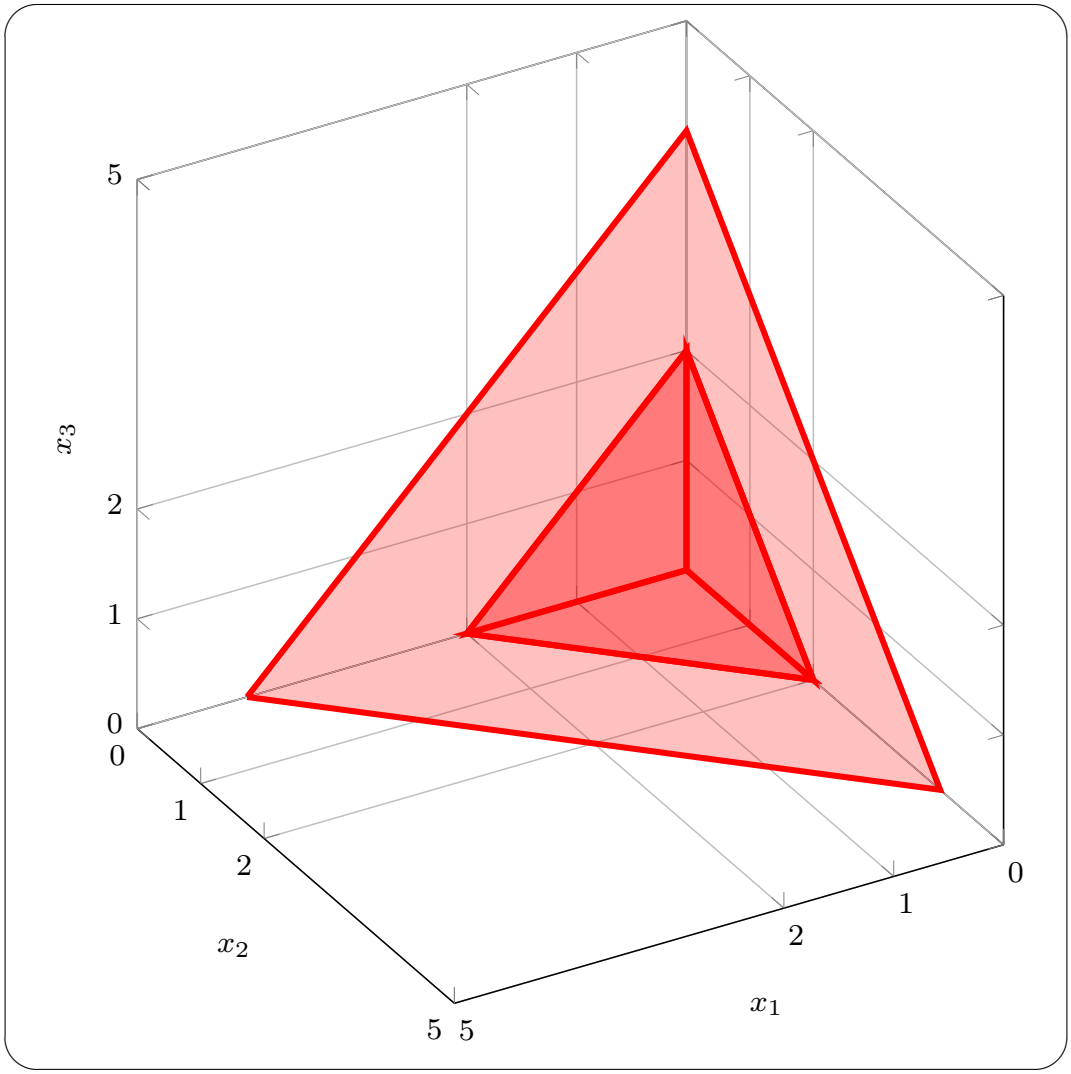
8.1 LP Problem *P81*

$$
\begin{array}{rrcrcrcl}
\max & & x_1 & + & x_2 & + & x_3 & \\
\text{subject to} & & x_1 & + & x_2 & & & \leq 2 \\
& - & x_1 & + & x_2 & & & \leq 1 \\
& & x_1, & & x_2, & & x_3 & \geq 0
\end{array}
$$



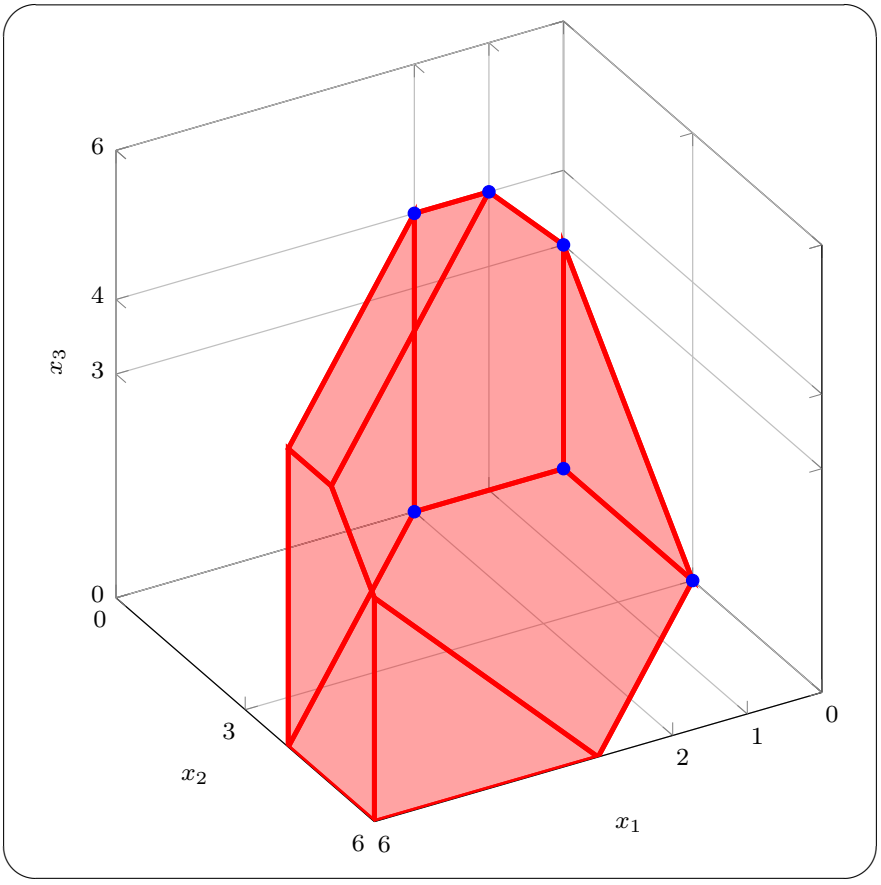
8.2 LP Problem *P82*

$$
\begin{array}{rrcrcrcr}
\max & & x_1 & + & x_2 & + & x_3 & \\
\text{subject to} & & x_1 & + & x_2 & + & x_3 & \leq & 2 \\
& - & x_1 & - & x_2 & - & x_3 & \leq & -4 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



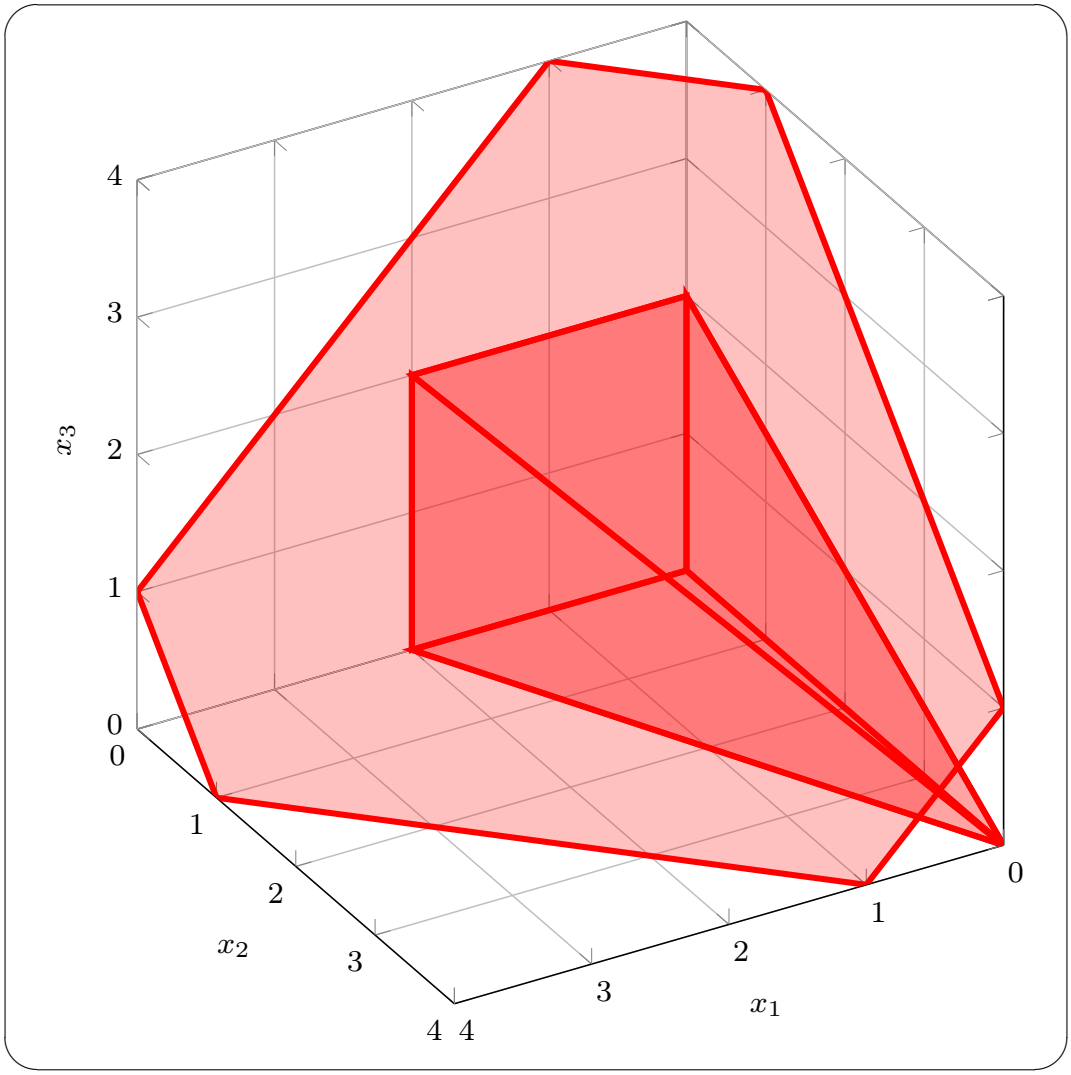
8.3 LP Problem *P83*

$$
\begin{array}{rrcrcrcl}
\max & x_1 & + & 2\,x_2 & + & x_3 & & \\
\text{subject to} & x_1 & - & x_2 & & & \leq & 2 \\
- & x_1 & + & x_2 & + & x_3 & \leq & 3 \\
 & & & & & x_3 & \leq & 4 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



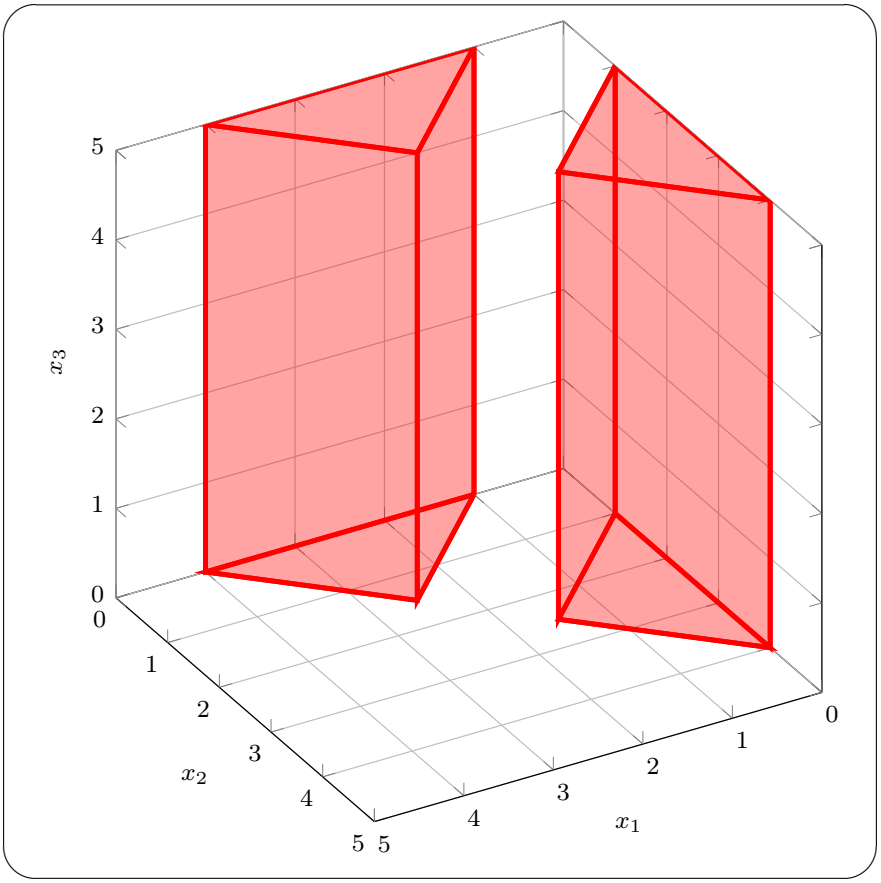
8.4 LP Problem *P84*

$$
\begin{array}{rrcrcrcr}
\max & x_1 & + & x_2 & + & x_3 & & \\
\text{subject to} & 2\,x_1 & + & x_2 & & & \leq & 4 \\
 & & & x_2 & + & 2\,x_3 & \leq & 4 \\
- & x_1 & - & x_2 & - & x_3 & \leq & -5 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



8.5 LP Problem *P85*

$$
\begin{array}{rrrrrrrrr}
\max & & x_1 & + & x_2 & + & x_3 & & \\
\text{subject to} & & x_1 & - & x_2 & & & \leq & -1 \\
& - & x_1 & + & x_2 & & & \leq & -1 \\
& & x_1 & + & x_2 & & & \leq & 4 \\
& & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$



8.6 LP Problem *P86*

$$
\begin{array}{rrcrcrcr}
\max & 2\,x_1 & + & 2\,x_2 & - & x_3 & & \\
\text{subject to} & x_1 & + & x_2 & & & \leq & 4 \\
 & x_1 & & & - & x_3 & \leq & 1 \\
- & x_1 & - & x_2 & & & \leq & -1 \\
 & x_1, & & x_2, & & x_3 & \geq & 0
\end{array}
$$

